TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models

Daniel Kushnir KUSHNIR.DANIEL05@GMAIL.COM
Hao Wang (corresponding author) HOGUEWANG@GMAIL.COM
Rutgers University

Abstract

Time-series foundation models (TSFMs) are used zero-shot, but the interpretability methods developed for language models do not transfer to them directly. Tokens cover different spans of time in different architectures, real series offer no ground truth for the concepts a probe might find, and the effects of single internal features on a forecast are small enough that uncontrolled measurements yield confident false positives. We present TSFM-Lens, an open-source system that compares TSFMs on a measured common time axis, validates its instruments on a benchmark with known generative components and on a forecaster with planted concepts, defines concepts by their effect on the forecast relative to matched random interventions, and tests exploratory claims once on a sealed held-out split. We apply it to seven TSFMs and confirm the results on two independent corpora. Once each model's tokens are concatenated in time order, all 21 model pairs share representational geometry above a shuffled-series null, confirmed on a fresh sealed split; the usual window average understates TimesFM's similarity to four fine-token models by 0.30–0.43 CKA. Different architectures also select the same input series for a concept: all registered transfer claims confirm on both private splits and on 800 raw real-world windows. Causal features act mainly on forecast level and seasonality, but fine-grained concepts are organized by model, and no concept spans more than three of the seven models. Activation prominence is a poor guide to causal importance, since 83.5% of prominent features have no held-out effect, and fewer than half of single-feature causal claims replicate. Against random directions drawn from the activation distribution, single features are only weakly specific (1.29 times chance on the larger corpus), and neither more data, a more powerful test nor gradient-selected candidates changes this. Every claim is reported with its evidence class and a reproduction pointer.

Keywords: mechanistic interpretability, time-series foundation models, sparse autoencoders, causal intervention, pre-registration

1 Introduction

Pretrained forecasting models are now default components. TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024), Moirai (Woo et al., 2024), MOMENT (Goswami et al., 2024), Lag-Llama (Rasul et al., 2023) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. Their architectures differ sharply: some are decoder-only stacks over 32-step patches, others are encoder–decoders over quantized single steps, encoders with cross-series attention, or decoders with sampled flow-matching heads. Leaderboards say which of them is more accurate. They do not say what these models represent, whether architectures this

©2026 Daniel Kushnir and Hao Wang.
License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/.

different represent the same things, or whether any internal structure we can name drives the forecast.

Mechanistic interpretability offers the instruments to ask: probes (Alain and Bengio, 2017; Hewitt and Liang, 2019), representational similarity (Kornblith et al., 2019), activation patching (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024), lenses (Belrose et al., 2023) and sparse dictionaries (Bricken et al., 2023; Gao et al., 2025), packaged in libraries such as TransformerLens (Nanda and Bloom, 2022) and pyvene (Wu et al., 2024). Early work applies pieces of this toolkit to single TSFMs (Wiliński et al., 2025; Mishra, 2026; Oublal et al., 2026; Divo et al., 2026). Three properties of forecasting models prevent a direct transfer, and each produces silent errors rather than crashes.

(i) **Tokens are not comparable across models.** One token covers 1 time step in Chronos-T5, 16 in Chronos-2, 32 in TimesFM and 96 in Timer. A cross-model comparison therefore needs a verified mapping from tokens to time.

(ii) **There is no ground truth for concepts.** Real series carry no labels for trend, period, changepoints or noise process, so a probe that "finds seasonality" cannot be checked on them.

(iii) **Effects are small and easy to fake.** Removing one sparse feature typically moves a forecast by about a tenth of the context's standard deviation, less than the dictionary's own reconstruction error. At this scale a precision mismatch, an intervention the forecast head never reads, or a null measured on the wrong rows each produce a clean, confident and wrong result.

We present TSFM-Lens, a system built around these three problems, and use it to study four research questions:

RQ1. Which forecast-relevant concepts do TSFMs learn?

RQ2. Are these concepts shared across architectures, in the sense of responding to the same inputs, having the same effect, or both?

RQ3. Do the concepts causally affect the forecast, and can their importance be read from activations?

RQ4. Where in depth do these computations happen, and how comparable is depth across architectures?

The system makes three methodological commitments. It is *controlled*: a synthetic benchmark with exact generative ground truth, and a forecaster with planted concepts, give every probe a case with a known answer. It is *causal*: a feature counts as a concept only if removing it changes the forecast more than removing a random direction of matched size on the same series, and reach controls show that the intervention landed. It is *confirmatory*: findings on the development (dev) split are frozen in a hash-pinned registry and tested once on a sealed private split. Measured *time alignment* makes architectures comparable, and the output is a single report in which every claim states its evidence class.

Contributions.

C1. A comparative interpretability system (Section 3): measured token-to-time alignment, a zero-code adapter path for new checkpoints, and a gated pipeline that

turns a list of checkpoints into one self-contained report, together with a benchmark generator with leakage-audited tiers and sealed, bit-exactly regenerable splits.

C2. A causal concept methodology (Section 4) that defines concepts by their effect on the forecast and separates two notions of sharing that are usually conflated: *same inputs* and *same effect on the same inputs*.

C3. Known-answer validation of the instruments (Section 6): every null and agreement test is held to a pass criterion fixed in advance on a forecaster with planted concepts and decoys, which rejected two plausible designs.

C4. A seven-model study with one-shot confirmation on two corpora (Sections 7 and 8), with a statistical protocol of series-level resampling, a multiplicity ledger and registered hypotheses (Section 5).

Main findings. We study TimesFM 2.5, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5-Base, Timer and Time-MoE, and re-test the registered claims on a second, three times larger corpus.

- **RQ1.** Sparse-dictionary features whose removal moves the forecast more than a random direction of the same size exist in six of the seven models. They act mostly on forecast level and seasonality and form a graded continuum rather than discrete units. Against random directions drawn from the layer's own activation distribution, however, they exceed chance only 1.29-fold. Under mean, resample and clean-activation ablation the effects keep their direction, and 79 of 91 features still clear the null under at least one of them.
- **RQ2.** Representational geometry is shared above chance for all 21 model pairs, which a registered test confirms on a fresh sealed split, provided windows concatenate their tokens: averaging them, the common practice, hides 0.30–0.43 of TimesFM's CKA with the fine-token models. Different architectures also *select the same series* for a concept: all registered transfer claims confirm held-out on both corpora and on raw real-world windows. The same broad effect types (raising or lowering the level, damping seasonality) recur across models, but fine-grained concepts are organized by model: no concept spans more than three of the seven models, which also holds held-out.
- **RQ3.** Activation prominence does not identify causal importance: most prominent features have no held-out causal effect. Fewer than half of the registered single-feature causal claims replicate, and neither more data, a more powerful test nor gradient-selected candidates improves this.
- **RQ4.** How TimesFM and Chronos-2 react to corruptions, layer by layer, replicates held-out for 6 of 9 corruptions on v2 and 4 of 9 on v3 (8 of 9 keep their sign), with level shift flipping sign on both; absolute depth locations are comparable only after accounting for how much of each model's compute is captured.

Table 8 lists every headline result with its evidence class and status.

Organization. Section 2 reviews related work. Section 3 describes the system, Section 4 the measurements, Section 5 the statistical protocol and Section 6 the known-answer

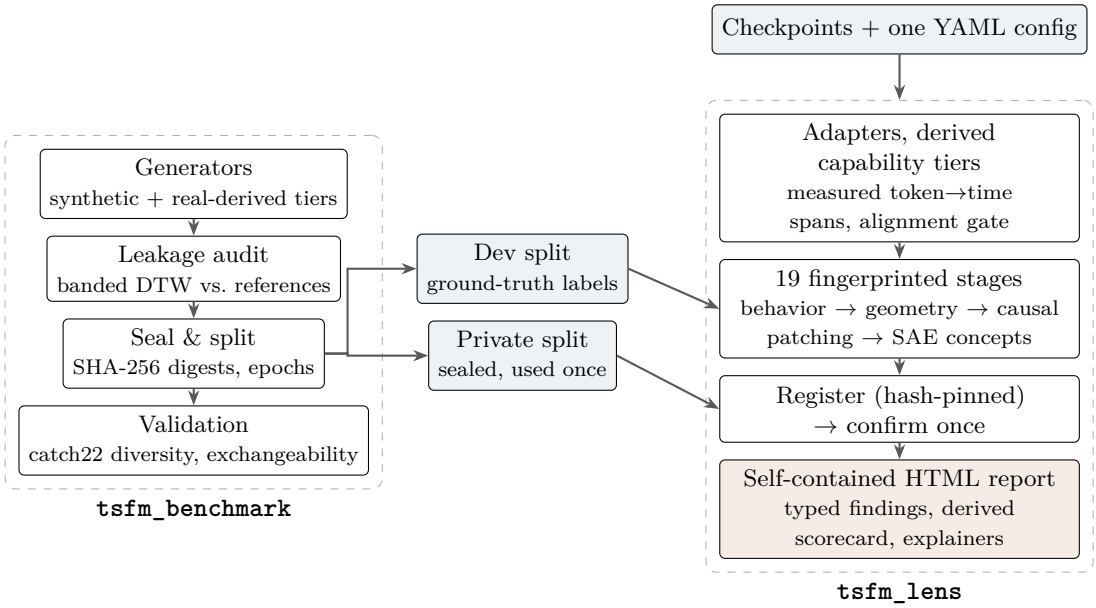


Figure 1: System overview. `tsfm_benchmark` produces leakage-audited, sealed corpora with exact ground truth; `tsfm_lens` turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by `confirm`, for claims registered in advance.

validation of the causal instruments. Section 7 gives the experimental setup and Section 8 the results. Section 9 discusses their implications and Section 10 their limitations. The appendices cover implementation and usage (Appendix A), additional validity analyses (Appendix B), statistical details (Appendix C), reproducibility (Appendix D) and an index of the public findings ledger (Appendix E). The ledger, `FINDINGS.md`, gives each result with exact numbers, evidence class, status and a reproduction command; identifiers such as MN-29 refer to its entries.

2 Related Work

Time-series foundation models. TimesFM (Das et al., 2024) is a decoder-only model over 32-step patches. Chronos (Ansari et al., 2024) quantizes values and trains a T5 encoder–decoder (Raffel et al., 2020) with a sampled decoder; Chronos-Bolt replaces the decoder with a direct patch head, and Chronos-2 (Ansari et al., 2025) is an encoder with cross-series group attention. Sundial (Liu et al., 2025) samples from a flow-matching head (Lipman et al., 2023), Timer (Liu et al., 2024) uses 96-step tokens, Time-MoE (Shi et al., 2025) is a sparse mixture of experts over per-step tokens, and Lag-Llama (Rasul et al., 2023) reads lagged values. Most of these models normalize each series internally (Kim et al., 2022), a detail an adapter must reproduce when it calls a model below its public API. Synthetic pretraining and evaluation data are common in this literature (Kang et al., 2020; Dooley et al., 2023); we use synthetic data instead to obtain ground truth for interpretation. Wan

et al. (2026) document flat, "collapsed" forecasts on weakly predictable targets, which we also observe (PM-07).

Interpreting time-series models. Saliency methods for time-series classifiers are unreliable across architectures (Ismail et al., 2020), and Kalnāre et al. (2025) apply patching and SAEs to time-series classifiers. For TSFMs, Wiliński et al. (2025) study representational redundancy and concept steering, and four recent studies train SAEs on a single TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026). Each examines one model or one method. None compares architectures on a common time axis, validates its probes against generative ground truth, separates input agreement from effect agreement, or tests its findings on held-out data.

Validity of interpretability methods. Probes need control tasks to separate what a representation encodes from what the probe learns (Hewitt and Liang, 2019; Belinkov, 2022). Activation patching is sensitive to the choice of metric and corruption (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024), subspace interventions can create illusory mechanisms (Makelov et al., 2024), and the apparent meaning of a direction can depend on the data used to inspect it (Bolukbasi et al., 2021). Sanity checks show that attributions can be insensitive to the model they explain (Adebayo et al., 2018). Attention weights are not explanations (Jain and Wallace, 2019). Sparse autoencoders are motivated by superposition (Elhage et al., 2022; Huben et al., 2024; Templeton et al., 2024), but their features vary across seeds (Paulo and Belrose, 2025), and crosscoders have been proposed to share one dictionary across models (Lindsey et al., 2024). Causal abstraction (Geiger et al., 2021) formalizes when an intervention supports a mechanistic claim. Our known-answer tests address the same concern empirically: each instrument must recover a planted answer before it is applied to a real model.

Comparing representations. Representational similarity (Raghu et al., 2017; Morcos et al., 2018; Kornblith et al., 2019; Kriegeskorte et al., 2008) and model stitching (Lenc and Vedaldi, 2015; Bansal et al., 2021) underlie claims that different networks converge on similar representations (Li et al., 2016; Huh et al., 2024). We use these measures only against shuffled-series nulls and show that, for TSFMs, they disagree with one another and with causal measures.

Tooling and research practice. Libraries such as TransformerLens, nnsight (Fiotto-Kaufman et al., 2025), pyvene, SAELens (Bloom et al., 2024) and Captum (Kokhlikyan et al., 2020) provide primitives for one model, but not comparative designs, nulls, multiplicity control or confirmation (Table 1). From empirical science we adopt pre-registration (Nosek et al., 2018), protection against adaptive reuse of holdouts (Dwork et al., 2015) and against analytic flexibility (Gelman and Loken, 2014; Simonsohn et al., 2020), leakage auditing (Kapoor and Narayanan, 2023) and series-level resampling (Efron and Tibshirani, 1993; Cameron et al., 2008).

3 The TSFM-Lens System

TSFM-Lens consists of two separately installable packages (Figure 1). `tsfm_benchmark` generates, audits, seals and validates corpora; `tsfm_lens` loads checkpoints through adapters,

Capability	*TransformerLens*	*nnsight / pyvene*	*SAELens*	*Captum*	*TSFM SAE studies*[†]	***TSFM-Lens***
Hooks, caching, activation patching	✓	✓	○	○	○	✓
Sparse dictionaries and feature ablation	○	○	✓	✗	✓	✓
Time-series models without per-model engineering	✗	○	✗	○	✗	✓
Measured token→time alignment across architectures	✗	✗	✗	✗	✗	✓
Benchmark with exact generative ground truth, leakage audit	✗	✗	✗	✗	○	✓
Random-direction nulls and reach controls for every ablation	✗	✗	✗	✗	✗	✓
Cross-model concept matching (inputs vs. effects)	✗	✗	✗	✗	✗	✓
Series-level resampling and multiplicity ledger	✗	✗	✗	✗	✗	✓
Registered, one-shot confirmation on sealed data	✗	✗	✗	✗	✗	✓
Typed evidence class and generated caveat per claim	✗	✗	✗	✗	✗	✓
One-command report designed for non-experts	✗	✗	○	✗	✗	✓

Table 1: Capabilities for comparative TSFM interpretability. ✓ provided; ○ possible with user code or partial; ✗ not provided. General libraries are single-model tools, complementary to TSFM-Lens, which uses the same primitives. [†]Mishra (2026); Oublal et al. (2026); Divo et al. (2026); Yıldırım (2026), judged from the papers.

runs a gated pipeline of analysis stages and renders the report. This section describes the parts that make cross-model measurements valid; implementation details and usage are in Appendix A.

3.1 Design principles

Each principle below is enforced in code as a gate, a refusal or a test, so that a measurement that cannot be trusted is withheld with a stated reason rather than reported.

P1. Ground truth by construction. Every probe is first validated on a case whose answer is known.

P2. Alignment is measured, not declared. Token spans are measured from impulse responses, and cross-model analysis happens only on a common time axis.

P3. Causal claims need a null, a reach control and a same-seed baseline. Every intervention is compared with a random intervention of matched size on the same series, and sampled models are seeded identically in compared calls.

P4. The series is the unit of inference. All intervals and p-values resample whole series.

P5. Explore, then confirm once. Dev results are exploratory; registered claims are tested once on a sealed split.

P6. Every claim is typed, and failures are loud. Each finding carries an evidence class and a generated caveat, and unsupported analyses are skipped with a rendered reason.

The evidence classes are BEHAVIORAL (input–output only), DESCRIPTIVE, ILLUSTRATIVE, GEOMETRIC (representational similarity), LINEAR-TRANSLATABLE (one representation linearly predicts another beyond an input baseline), CAUSAL WITHIN-MODEL (an intervention changes the forecast beyond a null) and HELD-OUT CONFIRMED (a registered claim replicated on the sealed split). A PREDICTIVE class marks internals that forecast a model's error without being shown to cause it.

3.2 Controlled benchmark generation

Real data enter the benchmark only as leakage references, for realism calibration and as raw material for explicitly labeled generators. Generators come in two tiers. The *synthetic* tier touches no real data and has exact ground truth. Its `parametric` generator is additive (trend, multiple seasonalities, AR noise, changepoints, anomalies, random walk, intermittency and heteroskedasticity), and `random_parametric` draws a recipe from one of eight frozen archetypes. The *real-derived* tier (`mixture`, `block_bootstrap` (Künsch, 1989) and `sequential_par`) inherits its source distribution and has only attributional ground truth. Because "free of leakage" and "realistic" pull in opposite directions, every sample carries its tier label and every analysis can be split by tier.

Every sample passes a two-stage leakage gate against the Monash archive (Godahewa et al., 2021), a cheap prefilter followed by banded DTW (Sakoe and Chiba, 1978), and a cross-split near-duplicate check. Dev and private splits use disjoint seed ranges mixed with SHA-256. Every sample is hashed, so that loading a sealed split with verification detects any change, and fresh private *epochs* are minted by striding the seed space. A validation package measures redundancy by shape matching and diversity in the catch22 feature space (Lubba et al., 2019), using the participation-ratio dimension (Gao et al., 2017) and nearest-neighbor tails rather than a 2-D embedding. It also tests dev/private exchangeability with an energy-distance test (Székely and Rizzo, 2013) and per-feature TOST equivalence (Schuirmann, 1987).

Two perturbation families are built in. Nine structured *corruptions* (noise, detrend, deseasonalize, frequency shift, level shift, spike, smooth, warp and dropout) drive the perturbation analysis of Section 4.1. *Generator counterfactuals* move one generative parameter along a dose ladder with every random draw held fixed; dose 1.0 must reproduce the original series bit-exactly, an identity test that caught a real replay bug (BM-05).

3.3 Model adapters and capability tiers

A model enters through an adapter that exposes loading, a `predict` method returning point and quantile forecasts in the original scale, the underlying module, input preparation, a forward pass and the time span of each token. The adapter's *capability tier* is derived from the methods it implements rather than declared: 0 (black box), 1 (observable hidden states with token spans), 2 (steerable, so patching is possible) and 3 (decomposable attention). Stages above a model's tier are skipped with the missing method named in the report. A

missing capability is a fact about the adapter; a measured failure of time localization is a fact about the model, and routes it to behavioral analysis only.

New checkpoints can use a zero-code adapter for Hugging Face models, which discovers layers, measures token spans and refuses rather than guesses. Its span discovery reproduces the spans of all six hand-written adapters exactly and recovers Timer's 96-step tokens from measurement alone (MP-01). Two of the seven models studied here use it.

3.4 Time alignment across architectures

A token index means a different stretch of time in each model. A 480-step context is 480 tokens in Chronos-T5 and Time-MoE, 30 in Chronos-2, Sundial and Chronos-Bolt, 15 in TimesFM and 5 in Timer, so "token 7" of two models covers different inputs. Raw token positions therefore cannot be compared across models. We first measure each token's time span I_j: an impulse is injected at every time step, and a token's span is the contiguous set of steps it responds to most, accepted only if it passes a peak-to-pedestal contrast gate and a contiguity gate (Appendix B). We then place every model's tokens on one shared grid of windows $W_w = [wL, (w+1)L)$, with $L = 32$ steps by default, by overlap-weighted averaging:

$$P_{wj} = \frac{|I_j \cap W_w|}{\sum_{j'} |I_{j'} \cap W_w|}, \qquad \tilde{h}_w = \sum_j P_{wj}\, h_j. \tag{1}$$

Here h_j is the activation vector of token j in one layer, and P_{wj} is the share of window w that token j covers. Each window's vector is a weighted average of the tokens that overlap it, in the model's own activation space. Three cases cover all seven models (Figure 2):

- **Tokens shorter than a window are averaged.** The 32 one-step tokens of Chronos-T5 inside a window receive weight 1/32 each, and the two 16-step tokens of Chronos-2, Sundial and Chronos-Bolt receive 1/2 each.
- **Tokens as long as a window map one to one.** Each 32-step TimesFM token becomes one window unchanged.
- **Tokens longer than a window are copied, not split.** A 96-step Timer token is the only token overlapping each of the three windows it spans, so all three windows receive the token's full vector with weight 1.

A token that straddles a window boundary contributes to both windows in proportion to its overlap with each. No token is ever cut into parts, because an activation vector has no internal time axis to cut along. A window that no token covers raises an error rather than becoming a zero row. A series-level vector is the mean of its window vectors.

What is compared across models. Pooling aligns the *rows* of two models' activations, not their *coordinates*. After pooling, row (s, w) of every model describes the same series s over the same time steps W_w, but the vectors still live in each model's own space: widths differ, from 384 in Time-MoE to 1,280 in TimesFM, and no axis of one model corresponds to an axis of another. No analysis therefore compares a vector of one model with a vector of another directly. Every cross-model measure compares something defined on the shared rows:

- **Geometry** (CKA and representational similarity analysis, RSA) compares how the rows relate to *one another* within each model. Two models score high when the series and windows that look similar to one model also look similar to the other. This depends on what a window's vector keeps, which we test below.
- **Stitching** fits a linear map from one model's rows to the other's on training series and scores it on held-out series. It reports only the gain over a map from simple input features.
- **Concept transfer** compares which *series* a feature fires on, a set of series identities that means the same thing in every model.
- **Causal effects** are measured on the forecast, which is in the time domain for every model. Interventions act on each model's own tokens, and patching never moves activations between models.

Is averaging fair? A test. Averaging can discard what distinguishes the tokens inside a window, and only models with tokens shorter than a window are averaged. We tested this directly on four disjoint sets of v3 development series: a 128-series pilot for development, then three sets of 512 for the main run, confirmation and validation. All criteria were fixed before each set was used (MN-38).

- **Averaging removes within-window order.** At the first captured layer, 40–71% of token-level variance in the 16-step and 1-step models lies within windows. Deeper layers fall to 14–33%, except Chronos-T5-Base, which stays at 70–76% at every depth. Pooled vectors still decode a window's level and spread, but not its slope or shape. In the first block of Chronos-T5-Base the slope decodes with $R^2 = 0.13$ from the window mean and 0.96 from finer sub-windows. TimesFM's unpooled token keeps this shape.
- **This biases cross-model geometry.** On identical rows, averaging leaves window-level CKA unchanged (within 0.05, layer ranking $\rho \geq 0.9$) for only 10 of 21 model pairs. The input itself shows the mechanism: its CKA with TimesFM's first layer is 0.15 when windows are averaged, 0.43 with four 8-step means and 0.54 with all 32 values.
- **Concatenation is lossless.** The fix is to concatenate a window's tokens in time order instead of averaging them, $\tilde{h}_w^{\text{cat}} = [h_j]_{I_j \subseteq W_w}$. This has no free weights. Linear CKA needs only the row Gram matrix, and the Gram matrix of a concatenation is the sum of the per-position Gram matrices, so the long vector is never formed. Concatenating 8-step means is not enough for 1-step models: Chronos-T5-Base's peak CKA with TimesFM keeps rising from 0.53 to 0.62 as the sub-windows shrink to single steps. A rule that adds a within-window trend to the mean restores decoding, but its CKA depends on how the two parts are weighted, so it is not used.

On a fresh set of 512 series with all captured layers, concatenation raises TimesFM's peak CKA with Chronos-2 from 0.45 to 0.73, with Chronos-Bolt from 0.39 to 0.65, with Chronos-T5-Base from 0.31 to 0.64 and with Sundial from 0.50 to 0.64, mostly at TimesFM's first layers. Pairs with Timer fall, by up to 0.10, because the other model's added order has no counterpart in Timer's copied token. Across the 21 pairs the two rankings correlate at $\rho = 0.80$. Window-level similarity in Section 8.3 is therefore reported with concatenated

windows on both corpora, and confirmed on a fresh private split. Concatenation is the tool's default (`alignment.window_rep: concat`); the mean remains available for comparison with earlier runs.

What the window rule affects. The window rule matters only where window vectors of two models are compared with each other: the window-level similarity measures, CKA and stitching; RSA compares whole-series summaries and is unaffected. Both window-level measures were re-measured. For stitching, concatenating the source window raises the gain over the input baseline from 0.498 to 0.554 on average over the 21 peak pairs in both directions, most when TimesFM is predicted (from Chronos-T5-Base 0.307 to 0.788), while the ranking of model pairs is kept (Spearman 0.92; MN-38). The stitching target stays the window summary, because a 1-step model's concatenated window has 24,576 coordinates and is underdetermined at the available number of series. The sparse-dictionary results (Section 4) do not depend on the rule, for three reasons.

- **No cross-model vector comparison.** An SAE is fit to one model's activations, and its features are compared across models only through quantities defined outside activation space: *which series* a feature fires on, and *how the forecast changes* when it is removed. A series is the same series for every model, and a forecast is a time series in every model's output, so neither quantity depends on how a window is summarized.
- **Series-level codes are symmetric.** The series a feature fires on are ranked by its series-level code, the mean over all 15 windows. This averages the whole context for every model alike, so it carries none of the within-window asymmetry that biases window-level CKA.
- **Interventions act on tokens.** Ablations encode and decode each token separately, and the forecast is measured on the model's own tokens, never on windows. An SAE is admitted only if its token-level reconstruction preserves the forecast. This check is needed because a window-level check is unreliable for fine-token models (Chronos-T5-Base ΔMASE +3.8972 at window level against +0.2457 at token level; MN-16).

Pooling does enter one step of the SAE pipeline: dictionaries are trained on window vectors, so a 1-step model's dictionary is learned from averages of 32 tokens and cannot represent within-window shape as a separate feature. This limits what a fine-token model's dictionary can express, not how dictionaries are compared, and the token-level admission check bounds its cost to the forecast. CONCAT cannot replace it, because a concatenated window of a 1-step model is a 32×768 vector with position-specific coordinates. Training on individual tokens is the alternative, and we tested it on 512 v3 series (3 seeds per arm, matched training steps, held-out series; MP-11). For TimesFM, whose tokens are windows, the two training sets are identical and no metric changed beyond seed noise. For Chronos-T5-Base, token training lowered token-level unexplained variance from 0.2835 to 0.0442 (block 3) and from 0.5469 to 0.1173 (block 11), cut the forecast change of the patched reconstruction from 1.020 to 0.113 MASE at block 11, and reduced dead features. Comparison remained defined in the same way, but the token-trained features of Chronos-Bolt shared fewer top series with TimesFM's (Chronos-T5-Base's change was within seed noise), and within-window slope was not more decodable from their series codes. The reported runs use window training.

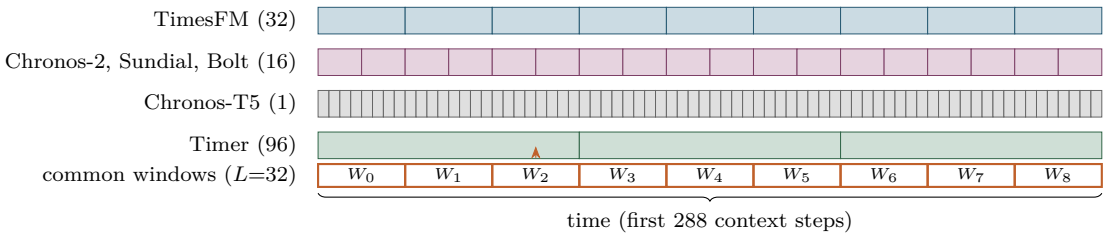


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5's one-step tokens are drawn in groups of four). Equation (1) pools them onto common 32-step windows: shorter tokens are averaged within a window, TimesFM's tokens map one to one, and each Timer token is copied to the three windows it spans.

Two further costs of the shared grid are not repaired by any window rule. First, the grid is no finer than the coarsest token. Within a 96-step span Timer has one vector, so the three windows it covers are identical, and an alignment check can attribute an impulse to the right window at most one time in three; alignment hits are therefore normalized by this attainable ceiling (MP-02). Analyses of lag structure use each model's matched resolution instead of the shared grid. Second, $L = 32$ is a convention. Re-pooling the stored windows to 64 and 96 steps leaves concept transfer and the size of geometric similarity almost unchanged, but moves the location of the peak layer pair for 12 of 21 model pairs (SH-23), so we report peak locations as window-dependent.

During extraction an *alignment gate* checks the pooling empirically. It injects an impulse into each window and checks, layer by layer, that the matching pooled window responds most, with the impulse amplitude calibrated per checkpoint and context length. Hits are compared with the attainable ceiling above. Deep layers of some models lose time localization, which the report shows and which qualifies window-level claims about those layers.

3.5 The stage pipeline and the report

An analysis is a pipeline of 19 stages that write typed artifacts (Figure 3). Each stage declares the configuration keys it depends on, and these are fingerprinted, so a changed setting invalidates exactly the dependent cached artifacts. Three gates record their decisions and the report names them as skip reasons: the *tier* gate, the *run-shape* gate, which drops comparison stages when only one model is given, and the *routing* gate for models that are not time-localized. Every intervention goes through a single patching primitive, which raises an error if the model silently splits a batch.

The deliverable is one self-contained HTML file, designed so that a reader without interpretability training can follow it while an expert can audit every number (Figure 4). Its sections pose plain questions, such as "Is the comparison fair?" or "What causally drives the forecast?". Each stage opens with a card stating its question, its method, what good and bad results look like and what it cannot show, and every figure carries a caption and an expandable explanation. The summary scorecard is computed from the artifacts by printed rules, with no model names hard-coded: each row shows a value, the reference it is compared with (a null, floor, ceiling or baseline) and the rule that yields the verdict. Every claim is

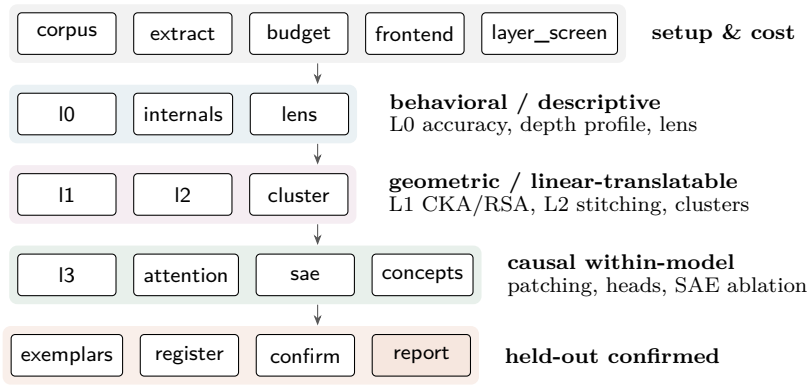


Figure 3: The 19-stage pipeline, grouped by the strongest evidence class each group can produce. Gates (tier, run shape, routing, configuration fingerprints) skip a stage per model with a rendered reason; only `confirm`, on the sealed private split, produces HELD-OUT CONFIRMED evidence.

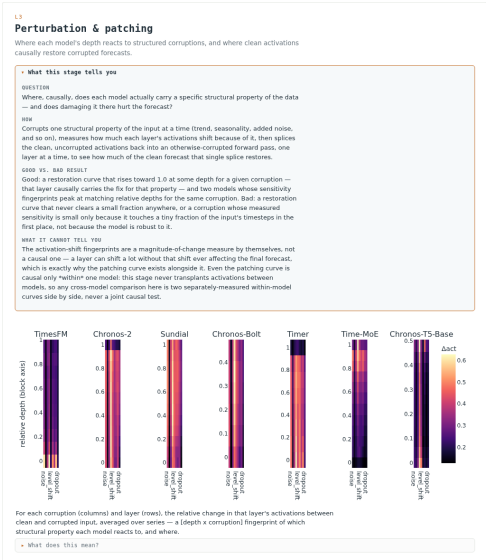


Figure 4: Excerpt of the generated report (perturbation section of the seven-model example report, `examples/panel7_v2`). The stage card states the question, method, how to read good and bad results and the limitation, above the per-model corruption-sensitivity fingerprints.

a typed record with its evidence class, a plain-language sentence and a generated caveat. Skipped stages show the gate's reason, failed sections render visibly, and a multiplicity ledger flags correction families that cannot reach significance.

4 Measurement Methods

Throughout, $x \in \mathbb{R}^T$ is a context, $y \in \mathbb{R}^H$ its continuation, $\hat{y} = f(x)$ a point forecast and $h^{(\ell)}$ the residual stream after block ℓ. The analyses are organized in levels: L0 (behavior),

L1 (geometry), L2 (linear translatability) and L3 (perturbation and patching), followed by depth analysis and causal concepts.

4.1 Behavior, geometry and perturbation (L0–L3)

L0: behavior. Accuracy is the per-series MASE (Hyndman and Koehler, 2006), with the seasonal lag taken from ground truth and a floor on the naive denominator; without the floor, intermittent series spuriously improve (MN-23). sMAPE, pinball loss (Gneiting and Raftery, 2007) and interval calibration are reported alongside. Model pairs are compared per generator family with a paired series bootstrap, Holm-corrected across all pairs and families (Holm, 1979), and differences are read against each model's repeat-run noise floor. Two further stages measure parameters, FLOPs, latency and the share of a forecast's FLOPs spent in captured blocks, and test what each model's input pipeline does before any layer runs (scale equivariance, handling of missing values, quantization).

L1 and L2: geometry. For aligned and centered window representations A and B over the same series, linear CKA (Kornblith et al., 2019), $\mathrm{CKA}(A, B) = \|B^\top A\|_F^2 / (\|A^\top A\|_F \|B^\top B\|_F)$, is computed for every layer pair of every model pair. A window is represented by its tokens concatenated in time order (Section 3.4), so CKA is computed from the row Gram matrices $K = \sum_p X_p X_p^\top$ summed over token positions p. Each value carries a series-level cluster bootstrap and a null that permutes whole series (1,000 permutations). L2 fits ridge maps between layers (Bansal et al., 2021) and reports only the *gain* over a baseline that predicts the target layer from input features, since much of any layer is linearly predictable from the input alone. Agreement between activation clusterings is measured by adjusted mutual information (Vinh et al., 2010).

L3: perturbation. For each corruption c, the sensitivity *fingerprint* of a model is the relative activation change $\|h^{(\ell)}(c(x)) - h^{(\ell)}(x)\| / \|h^{(\ell)}(x)\|$ as a function of layer, and its *restoration curve* is the fraction of forecast damage undone by patching clean activations of layer ℓ, whole or per window, into the corrupted pass. Two models are compared by the rank correlation of their fingerprints on a shared relative-depth axis, never by transplanting activations between models. Restoration is causal within a model; agreement between two models' fingerprints is correlational.

4.2 Depth

A skip lens decodes the forecast from each layer through the model's own head, and a tuned ridge lens (Belrose et al., 2023) fits a readout per layer. The depth axis is the fraction of the *whole* stack, including parts that are not captured: Chronos-T5's decoder is not captured, so its last captured encoder block sits at depth 0.478 (DE-04). The lens is withheld for a model whose head reads positions that patching never writes, where it would return a flat and meaningless curve (DE-03).

4.3 Causal concepts

Sparse dictionaries. Each target layer receives a TopK sparse autoencoder (SAE) (Makhzani and Frey, 2013; Gao et al., 2025),

$$z(h) = \mathrm{TopK}\left(W_e(h - b_d) + b_e\right), \qquad \hat{h} = W_d\, z(h) + b_d, \tag{2}$$

with a searched dictionary size, an auxiliary loss that revives dead atoms and a minimum number of training steps (MP-03). Because single SAE seeds are not reproducible (Paulo and Belrose, 2025), three replicate dictionaries per target provide a within-model reproducibility ceiling (MP-04).

The ablation battery. Feature i is tested on the k series on which it fires most strongly. The forecast under the reconstruction with $z_i = 0$, written $\hat{h}_{-i}$, is compared with the forecast under the full reconstruction:

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \tag{3}$$

where g_c is one of nine forecast *channels*: trend, seasonal strength, spectral centroid, level, dispersion, near- and far-horizon shape distance, MASE and flatness. Using the reconstruction as the baseline keeps the SAE's reconstruction error in both arms. This matters because that error moves the forecast several times more than a typical feature does (Section 8.4). The two shape channels are distances and are therefore treated as unsigned.

Random-direction nulls. The null asks whether removing the feature's direction matters more than removing the same amount of activation along another direction. Two variants are used.

- The *profile-matched* null (used for the v2 corpus) removes, on the same rows and at every token t, exactly the amount the feature removes, $|z_i(t)|\,\|w_i\|$, along each of 16 random *decoded* directions: a unit vector drawn isotropically in the dictionary's code space and mapped through the decoder. Only the direction differs from the real ablation. We call this null isotropic for short.
- The *covariance-matched* null (used for the v3 corpus) is identical except for the direction, which is a random Gaussian combination of the same rows' centered clean token activations. It is thus a draw from a Gaussian with the layer's sample activation covariance and lies in the subspace the model actually uses. It is the stricter of the two: on v3 it reduces the clearing cells from 4.92 times the empirical chance rate to 1.27 times (Appendix B).

Feature i *clears* channel c if its mean $|\Delta_i^{(c)}|$ exceeds the 95th percentile $q_{95}^{(c)}$ of the pooled null, and its effect vector is $v_i^{(c)} = \overline{\Delta_i^{(c)}}/q_{95}^{(c)}$, the signed effect in units of its own null. Because a mean over rows is compared with a percentile of single rows, the chance rate of clearing is not 5%. It is estimated per cell by scoring each null draw against the others (leave-one-draw-out), and ranged from 0.3% to 4% per target in the case study. Matching the removed amount token by token is what lets the null pass the known-answer test of Section 6; a simpler null that removes one uniform, batch-averaged amount fails it (MN-29).

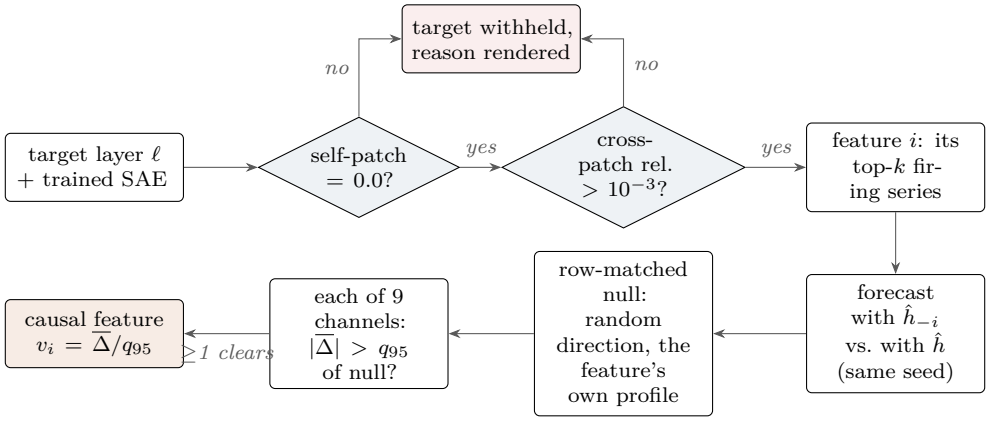


Figure 5: The ablation battery. A target is scored only after both reach controls pass; each feature is compared, channel by channel, with a random-direction null on exactly its own top-firing series.

Reach controls. Before any feature is scored, a *reach probe* requires that patching a layer's own activations back into it changes the forecast by exactly 0.0, and that patching another layer's activations changes it by a relative amount above 10^{-3}. A target that fails either check is withheld with its reason (Figure 5).

From features to concepts. Causal features are grouped at three granularities.

- *Per-target concepts* cluster one layer's effect vectors with k-means, choosing k by silhouette (Rousseeuw, 1987) among values whose smallest cluster has at least three members. Without this constraint the best silhouettes are one-outlier splits (SH-20). A target with no admissible k is called *non-modular*.
- The *atlas* pools features from all models and layers and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a tight, shared causal-effect profile.
- *Families* are a coarser tiling for reading (average linkage at cosine 0.5), tested against a column-shuffle null and titled deterministically from their channels.

4.4 Two notions of sharing

Whether a concept is "shared" between two models is two separate questions.

Same inputs. *Concept transfer* asks whether some feature in another model separates a concept's top-k series from the rest, measured by AUC, against stratum-matched random subsets of series. It is run in both directions; a transfer is *reciprocal* if both directions clear, with Benjamini–Hochberg control (Benjamini and Hochberg, 1995) per ordered model pair and direction. Input agreement between two concept parts is measured by the overlap of their top-k series (hypergeometric test), never by whole-series correlation: two sparse features can correlate at $\rho = 0.80$ while sharing no top series (MN-19).

Same effect on the same inputs. For every reciprocal transfer, both sides are ablated on the same series set U. Each side must first clear its own null on U. The two sides then

agree if their level concordance and shape cosine both exceed the floors given by ablating random feature sets matched to each side's activation. A statistic below both floors' 5th percentile is called *acts differently*; one that merely fails to exceed the 95th percentile is *no specific agreement*. The distinction matters: reading the second case as the first once labeled 109 of 288 tests as disagreements, against 9 under the correct test (MN-18).

Evidence ladder and sharing classes. The report summarizes each atlas concept on a six-rung ladder: R1 same forecast effect (atlas membership), R2 same inputs, R3 reproducible across SAE seeds, R4 other dictionaries select the same inputs, R5 same causal effect on the same inputs, and R6 confirmed on private data (the report labels the rungs L1–L6). Each concept also receives a *sharing class*: *shared* (same effect and same inputs), *partially shared*, *convergent* (same effect reached on different inputs) or *single-model*. A concept part is flagged as provenance-driven when at least 80% of its top series come from one real-derived generator.

5 Statistical Protocol

Four mechanisms control how often a report with thousands of comparisons is fooled (Table 2).

The series is the resampling unit. Windows of a series share its trend, noise and scale, so every interval and p-value resamples whole series (Efron and Tibshirani, 1993; Cameron et al., 2008): paired bootstrap for accuracy, cluster bootstrap for CKA, validation-series intervals for stitching, paired cluster bootstrap for patching. Splits are by series and stratified, never the first rows of a corpus that is stored grouped by task.

A multiplicity ledger. Every test belongs to a named family, corrected by Holm for confirmatory families and by Benjamini–Hochberg for discovery-oriented atlas transfers. Resampled p-values are floored at $1/n_{\text{boot}}$ (Phipson and Smyth, 2010), so a Holm family of m tests cannot go below m/n_{boot}. The ledger computes this minimum for each family and flags families that can never reach significance, a problem that grows quadratically with the number of models compared. The seven-model run has 123 families with 4,238 comparisons; its strictest accuracy family (105 tests over 21 model pairs) has a minimum Holm p of 0.021 at $n_{\text{boot}} = 5000$. Two kinds of test are judged outside the ledger: the 8,705 feature-channel cells of the ablation battery are judged by their count against the empirical chance rate, and shared-input agreement tests by their own per-test floors.

References measured in the run itself. Every effect is read against a reference measured in the same run: repeat-run noise floors, shuffled-series nulls, input-feature baselines, random-direction nulls, activation-matched floors or replicate-seed ceilings. Before the private split is opened, `confirm` also simulates each claim's minimum detectable effect, for example a ΔMASE of 0.133 at $n = 290$ with 80% power, so that a non-replication can be read as either an absent effect or an underpowered test.

Register, then confirm once. The `register` stage freezes hypotheses from dev results together with the content hash of each source artifact, and hashes the registry. `confirm` refuses mismatched hashes, requires the private seal to verify, tests each claim once and refuses to overwrite its own artifact; a forced rerun is recorded as a `repeated_look`. A

Claim type	Test statistic	Null / reference	Correction	Held-out
Model A more accurate (per family)	paired ΔMASE, series bootstrap	zero difference; repeat-run noise floor	Holm over all pairs × families	✓
Layers share geometry	linear CKA, cluster bootstrap	shuffled-series null at the same layer pair	per model pair	✓
Layer linearly translatable	ridge stitching gain	input-feature baseline (gain = 0)	per pair	–
Depth reaction agrees	Spearman ρ of fingerprints on shared depth	bootstrap CI of ρ	per corruption	✓
Layer restores forecast	restoration fraction	self-patch = 0, cross-patch ≠ 0	per model	–
Feature is causal	mean $\lvert\Delta^{(c)}\rvert$ on own rows	matched random-direction null, q_{95}	per channel (count vs. chance); Holm when registered	✓
Concept reproducible	best-match cosine across seeds	within-model replicate ceiling	–	–
Same inputs	top-k AUC in other model	stratum-matched random subsets	BH per ordered pair and leg; Holm when registered	✓
Same effect on same inputs	level concordance; shape cosine	activation-matched feature-set floors (q_{95} agree, q_{05} differ)	per test; Holm when registered	✓
Concepts are cross-model	atlas centroid cosine and member pairs	column-shuffle and model-label nulls	Holm when registered	✓
Internals predict failure	gain in log-MASE Spearman over a free baseline	zero gain	Holm when registered	✓
Families are real	number and coverage of families	column-shuffle null	–	–

Table 2: Every claim type the case study reports, the statistic that tests it, the reference it must beat, its multiplicity correction and whether the `confirm` stage currently re-tests it on private data. Claim types without a held-out mark are reported as exploratory.

consumed split is replaced by a fresh epoch, which is checked for sample-hash overlap with every earlier split of every corpus version. This is pre-registration (Nosek et al., 2018) implemented in software, and it guards against adaptive reuse of the holdout (Dwork et al., 2015; Gelman and Loken, 2014). Findings that were not registered cannot be confirmed after the fact.

6 Known-Answer Validation of the Causal Instruments

Every causal count in Section 8 rests on a null, and a null can be wrong without raising any error. Before using an instrument on real models we therefore test it on a forecaster whose answer is known.

The planted forecaster. TSFM-Lens ships a mock forecaster, `mock_planted`. One block of it writes known unit directions into the residual stream on the series that have a given property. The properties are thresholded readouts of the patch mean, slope, residual deviation, seasonal projections and a spike statistic, each firing on 10–30% of the corpus. The forecast head then applies a known effect (on level, trend, seasonality or dispersion). The model also carries decoys that look like concepts: *input-only* directions with zero causal effect and *sub-null* directions with a weak effect. A control layer placed after the planted block has no planted effect. For the cross-model tests there is a pair of architectures (widths

Instrument	Result on the planted forecaster	Pass criterion	Verdict
Profile-matched null (v2)	false-positive rate $\leq$0.003, sensitivity 0.70–1.00, all 5 seeds	FPR $\leq$0.10, sens. $\geq$0.5	pass
Uniform batch-averaged null	false-positive rate 0.168–0.248 on every seed	FPR $\leq$0.10, sens. $\geq$0.5	fail
Covariance-matched null (v3)	sensitivity 0.916, false positives in 5 of 2,948 control cells	FPR $\leq$0.10, sens. $\geq$0.5	pass
Gradient-attribution candidate ranking	mean top-20 recall 0.831 on unseen seeds	recall $\geq$0.8	pass
Powered per-feature test (32 series, 64 null draws)	sensitivity 0.6, against 0.911 for the 8-series rule	sens. $\geq$0.8	fail
Shared-input agreement (R5)	false "same" 0 of 9 decoys; detects a shared one-direction concept in 0.4 of seeds, a 3–4-direction one in 0.0	–	specific, insensitive
Subspace redesign of R5	sensitivity 0.5 and 0.375; false agreement 0 of 14	sens. $\geq$0.8, false $\leq$0.05	fail

Table 3: Known-answer tests of the causal instruments on the planted forecaster. Each criterion was fixed before the test was run. Ledger entries: MN-29 (nulls), MN-30 and MN-32 (agreement), MN-36 (powered test), MN-37 (attribution).

64 and 96, depths 5 and 7, rotated residual bases) that share planted concepts, plus a twin with the opposite effect.

Protocol. A scorer reads only the planted manifest and the run's artifacts. The pass criteria were fixed before any data were seen and are listed per instrument in Table 3: for the nulls, a per-cell false-positive rate of at most 0.10 on the control layer and a sensitivity of at least 0.5; for designs proposed as replacements of a working instrument, a sensitivity of at least 0.8. Each is evaluated on five construction seeds. Seed 0 was used to tune the attribution rule and seeds 1–4 were held out. The criterion is evaluated at one effect size (dose 1).

Results. Table 3 lists every instrument that was held to such a test. Three results shape the rest of the paper.

First, the test has teeth. The profile-matched null passes on all five seeds, while a simpler null that removes one uniform, batch-averaged amount fails on every seed, because a strong or dense atom clears it on size alone. On seed 0 the profile-matched null also keeps every sub-null decoy from clearing, raises the adjusted Rand index of the atlas against the planted concepts from 0.336 to 0.469, and raises sharing-class accuracy from 0.364 to 0.714. Re-scoring a four-model development run under the profile-matched null reduced its causal features from 293 to 110 (MN-29). The profile-matched null may be conservative in the planted world, since a random direction overlaps the planted span more than a decoy does.

Second, the agreement rung R5 is specific but insensitive. It never calls a decoy "same", so its positive verdicts can be trusted, but it rarely detects a concept that two planted

Model	Checkpoint	Architecture	Token	Head	Tier	Params	FLOPs obs.
TimesFM 2.5	`google/timesfm-2.5-200m`	decoder-only	32	deterministic	3	231M	42.5%
Chronos-2	`amazon/chronos-2`	encoder (+group attn.)	16	deterministic	3	119M	97.5%
Sundial	`thuml/sundial-base-128m`	decoder + flow head	16	sampled	2	128M	21.3%
Chronos-Bolt	`amazon/chronos-bolt-small`	encoder + patch head	16	deterministic	2	48M	79.1%
Chronos-T5-Base	`amazon/chronos-t5-base`	encoder–decoder	1	sampled	3	201M	13.9%[a]
Timer	`thuml/timer-base-84m`	decoder-only	96	deterministic	2[b]	84M	99.8%
Time-MoE	`Maple728/TimeMoE-50M`	decoder-only, MoE	1	deterministic	2[b]	113M	50.0%

Table 4: The seven models. "Token" is time steps per token; "Tier" is the derived capability tier (Section 3.3); "FLOPs obs." is the share of a forecast's FLOPs spent in captured blocks at context 480, measured with PyTorch's FLOP counter. [a]Encoder only (DE-05). [b]Via the zero-code adapter. Adapters also exist for Lag-Llama, whose tokens read disjoint lags and therefore receive behavioral analysis only, and for other Chronos-T5 sizes (8M–709M).

architectures truly share, and quadrupling the number of shared series does not change this (MN-30). An absent "same" verdict on real models is therefore not evidence of a difference. A redesign that ablates a concept's member subspace on each side also fails its pre-fixed criterion. Its failure is a ceiling rather than a design detail: even the planted distributed subspace does not move the forecast more than an equal-sized random removal (MN-32).

Third, more statistical power per feature does not help. A test using four times as many series and null draws is *less* sensitive than the eight-series rule, because the null is common-mode across series and extra rows barely tighten it (MN-36).

Known-answer tests establish what an instrument can detect and how often it is fooled in a world whose answer is known. They do not establish that its verdicts on real models are correct.

7 Experimental Setup

Models. We analyze seven TSFMs (Table 4): TimesFM 2.5 (Das et al., 2024), Chronos-2 (Ansari et al., 2025), Sundial (Liu et al., 2025), Chronos-Bolt and Chronos-T5-Base (Ansari et al., 2024), Timer (Liu et al., 2024) and Time-MoE (Shi et al., 2025). Timer and Time-MoE are loaded through the zero-code adapter. Together the panel covers decoder-only, encoder-only and encoder–decoder stacks, token widths from 1 to 96 steps, a mixture-of-experts model and deterministic and sampled forecast heads.

Corpora. All claims are tested on two independently generated corpora, each with its own dev split, sealed private split and registry.

- **v2** (primary). The dev split has 965 series in five generator families; 410 of them are real-derived, built from Monash weather, ETT (Zhou et al., 2021) and the Chronos electricity and traffic collections. The private split is a sealed epoch of 971 series that shares no sample with any earlier split. Dev and private splits are exchangeable: an energy-distance test gives $p = 0.951$, 20 of 22 catch22 features are equivalent by TOST and the other two are inconclusive, and 6 of 937,015 cross-split pairs are near-duplicates.

- **v3** (replication, three times larger). The dev split has 2,876 series, 2,091 synthetic and 785 real-derived from 15 real sources, adding ERCOT load, M4 hourly, solar, taxi, KDD Cup air quality and pedestrian counts to the v2 sources. The private split is 1,024 series drawn from a sealed epoch of 2,868. An *external* set of 800 raw windows from 12 GIFT-Eval datasets (Aksu et al., 2024), cut without any generator and sharing no sample with either split, re-tests the registered transfer claims and is reported separately from the confirmation verdict. Exchangeability, computed after the confirmation look and without any model, gives energy-distance $p = 0.995$ and 22 of 22 features equivalent. A second sealed v3 epoch (2,863 series, no sample shared with any earlier split of any corpus) was minted later and opened once, for the registered window-geometry claims (Section 8.3).

Contexts have 480 steps, a multiple of Timer's 96-step token, the horizon is 64 steps and the alignment window 32 steps.

Protocol. On v2, SAEs are trained on 56 target layers (TimesFM 10, Chronos-2 9, Sundial 9, Time-MoE 9, Chronos-T5-Base 9, Timer 6 and Chronos-Bolt 4), each with three replicate seeds, and causal features are scored against the profile-matched null. On v3, SAEs are trained on 50 targets and every causal claim is scored and registered against the stricter covariance-matched null; every setting of the v3 run was fixed before its data existed. Bootstrap intervals use $n_{\text{boot}} = 5000$. The v2 registry holds 128 hypotheses, 82 of them replicable on the private split; the v3 registry holds 111, of which 63 are replicable. Each private split was opened once.

Sources of numbers. Unless stated otherwise, numbers are from the v2 dev run or its confirmation. Numbers from v3 are labeled as such. The v2 run used 21.1 hours of stage time on one RTX A5000 (24 GB); its artifacts are in `examples/panel7_v2/run`, and Appendix D gives the full provenance.

How the models behave on this corpus. Interpretability results are only meaningful if the models do the task they were trained for. Table 5 summarizes their behavior on the v2 dev split. Chronos-2 is the most accurate. Five models beat the seasonal-naive forecast on 76.5–84.5% of series, Timer on 69.5% and Time-MoE on only 37.8%; Time-MoE is less accurate than the naive forecast overall, so its internal results describe a model that does not forecast these series well. The nominal 80% intervals of Sundial and Chronos-T5-Base cover only 36% and 44% of outcomes, and four models propagate an injected missing value silently. On the private split, Chronos-2 is more accurate than TimesFM on the registered `mixture` family (ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$) and, with a smaller effect, overall (-0.070 [-0.129, -0.008], $p = 0.025$, $n = 971$). On v3 the `mixture` result replicates (0.303 [0.176, 0.445], $p_{\text{Holm}} = 0.0002$).

8 Results

We first summarize what replicated on held-out data (Section 8.1) and then answer the four research questions in turn. Each subsection opens with its answer, followed by the dev evidence and the held-out status on both corpora. Section 8.7 asks whether internals help

Model	median MASE	80% coverage	scale resid. (s.d.)	missing value	latency (ms)
Chronos-2	**1.2002**	0.757	0.0102	handled	23
TimesFM	1.2560	**0.779**	1.6×10^{-6}	propagates	39
Sundial	1.4131	0.360	0.0310	propagates	62
Chronos-T5-Base	1.4231	0.436	2.7×10^{-7}	handled	975
Chronos-Bolt	1.5487	0.732	0.0101	handled	15
Timer	1.9306	–	0.0262	propagates	5
Time-MoE	2.8642	–	0.0368	propagates	103
naive / seasonal naive	2.7593 / 1.9819				

Table 5: Behavior on the v2 dev split (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations, $c = 0.001$. Missing value: whether an injected NaN is handled or propagates silently. Latency: median time to forecast a batch of 8 series. Timer and Time-MoE output point forecasts only.

Registered claim type	v2 (971 series)	v3 (1,024 series)	Section
Accuracy (Chronos-2 better on `mixture`)	1/1	1/1	7
Peak-CKA pair	1/1	1/1	8.3
Concept transfer (shared inputs)	20/20	20/20	8.3
same claims on 800 raw real windows	–	20/20	8.3
Atlas structure	2/2	2/2	8.3, 8.4
Single-model atlas concept	1/1	–	8.3
Cross-model causal (atlas concepts; agreement)	0/3; 0/6	none registrable	8.3
Single-feature causal effect	10/31	11/24	8.4
Perturbation-profile agreement	6/9	4/9	8.6
Failure prediction from internals	2/2	2/3	8.7
Confirmed / replicable (not testable)	43/82 (6)	41/63 (3)	
Window geometry, second sealed v3 split: 21 pairs above null; mean understates 4 pairs	–	21/21; 4/4	8.3

Table 6: Held-out confirmation on two independent corpora. Each cell is confirmed / tested; each claim type is one Holm family. The external raw-window set and the second v3 split (2,863 series, of which 512 are used) are not counted in the totals. Sources: `confirm/confirmation.json` of the v2 and v3 runs (CA-11, CA-12).

predict forecast failure, and Table 8 at the end lists every headline result with its evidence class.

8.1 Held-out confirmation at a glance

Table 6 gives the outcome of the one-shot confirmations. On v2, 43 of 82 replicable claims confirm, 33 do not and 6 cannot be tested; on v3, 41 of 63 confirm and 3 cannot be tested. The pattern is the same on both corpora: claims about accuracy, geometry, shared inputs, the structure of the concept atlas and failure prediction replicate, fewer than half of the single-feature causal claims do, and no claim that two models share a causal effect replicates. One procedural event on v2, a crash before any statistic was written, is disclosed in Appendix D.

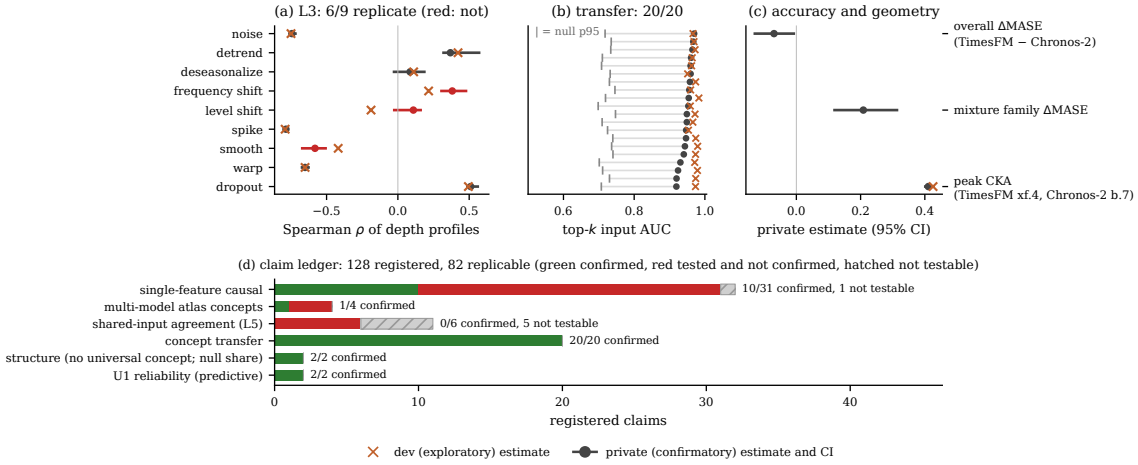


Figure 6: Claims re-tested on the sealed v2 private split (971 series; panel a uses 800 of them). (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2 (red: the dev value is not inside the private interval). (b) The 20 concept-transfer claims with private null 95th percentiles. (c) Accuracy and peak CKA; crosses are dev values, dots and bars private values with 95% series-bootstrap intervals. (d) Registered concept claims by type: confirmed, tested and not confirmed, and not testable (the one confirmed atlas claim is single-model). Source: `confirm/confirmation.json`.

8.2 RQ1: Which forecast-relevant concepts do TSFMs learn?

Answer. In six of the seven models, SAE features exist whose removal moves the forecast more than removing a random decoded direction of the same size. These features act mostly on forecast level and seasonality and form a graded continuum rather than discrete units. Against random directions drawn from the activation distribution itself, few single features stand out.

Causal features beyond isotropic directions. Of the 56 target layers, 55 passed both reach controls. The exception is Chronos-2's final block, where patching another layer moved the forecast by exactly 0, because its head reads placeholder tokens that a context patch never reaches (DE-03). Across the 55 targets, 442 feature-channel cells clear the profile-matched null where the empirical chance rate predicts 178.75, which is 2.47 times chance. The ratio ranges from 1.64 (Chronos-T5-Base) to 5.70 (Chronos-Bolt) for six models; for Timer it is 1.02, although its layers do reach the forecast (relative reach 0.066–0.261; PM-17). Tested target by target, 22 of 55 targets clear significantly more cells than chance (Benjamini–Hochberg at 0.05), including at least two in each model except Timer, which has none. At the level of whole features the excess is smaller: 194 features clear at least one channel, against at most 144.95 expected if channels were independent.

The effects concern level and seasonality. Nine effect families tile the causal features (Figure 7). The two largest act on level: *level raisers* (45 features) and *level lowerers* (44), which together hold 89 of the 192 assigned features. *Seasonality dampeners* (25) and

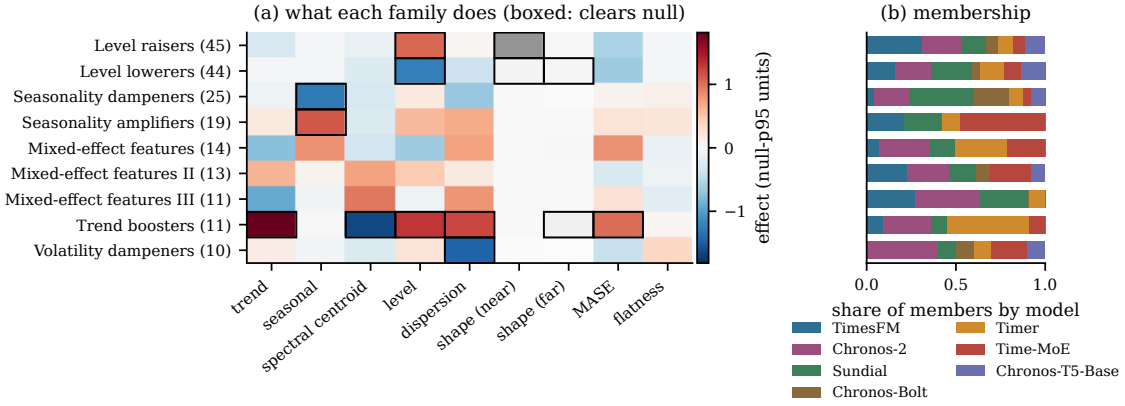


Figure 7: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature *does* (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

amplifiers (19) follow. Three mixed-effect families (14, 13 and 11 features), *trend boosters* (11) and *volatility dampeners* (10) are smaller.

The effects form a continuum. Three observations show that causal features are graded rather than organized into discrete units. First, within a layer, 50 of 56 targets are non-modular; the 10 per-target concepts that exist sit on 5 targets, and 4 of them are on Timer targets whose clear rate is not above chance. Second, the strict atlas assigns only 35 of the 194 features to its 11 concepts. This is more structure than a column-shuffle null produces (null 95th percentile of 5.05 concepts; assigned fraction 0.180 against 0.088; $p = 0.005$), but it covers few features. Third, the families cover 192 of 194 features but do *not* beat the shuffle null, either in their number ($p = 0.68$) or in their coverage ($p = 0.23$), and the silhouette is 0.25–0.32 for every cut from 6 to 16 families. The effect vectors are not low-dimensional (participation ratio 5.58 of 9 channels on v3; MN-07), so the absence of discrete families is not a lack of variation. The families are therefore a vocabulary for reading the effects, not discovered units.

Against on-manifold directions, features stand out only modestly. The v3 run scored every feature against both nulls. Under the isotropic null, 238 of 967 scorable features clear at least one channel. Under the covariance-matched null, whose random directions follow the activation distribution of the same rows, 91 do. The chance count must keep the dependence between the nine channels, because one random direction moves level and near-horizon shape together. We therefore score each null draw as a pseudo-feature on all channels at once, leaving that draw out of its own threshold. The expected counts are then 67.0 under the isotropic null and 70.4 under the covariance-matched null, so features exceed chance 3.55-fold against isotropic directions but only 1.29-fold against on-manifold ones (Poisson-binomial $p = 0.0045$; Table 7). Treating the channels as independent would put the on-manifold expectation at 111.2, above the observed count, which overstates chance. Despite the modest excess, 11 of the 24 registered single-feature claims replicate on new

Run	Null	Scorable	Clear ≥ 1	Expected	Ratio (p)
v3	isotropic	967	238	67.0	3.55 (4×10^{-86})
v3	covariance-matched	967	91	70.4	1.29 (0.0045)
v3$^+$	isotropic	1,118	304	75.9	4.01 (1×10^{-128})
v3$^+$	covariance-matched	1,118	112	80.3	1.40 (9×10^{-5})

Table 7: Features clearing at least one of nine channels against the chance count when channel dependence is kept (each null draw scored as a pseudo-feature on all channels, leave-one-draw-out). p is the Poisson-binomial upper tail over the per-feature chance rates. v3$^+$ is a separate scoring of the 38 differentiable v3 targets with gradient-nominated candidates added (Section 8.5), so its rows are not comparable with the 50-target v3 rows.

series (Section 8.4). The effects clearly exceed isotropic random directions but only modestly exceed typical on-manifold directions, a distinction we return to in Section 8.5.

Effects under on-manifold interventions. Zero ablation could move activations off the data manifold. We therefore re-ran the 91 v3 features that clear the on-manifold null under three further interventions, each with the null matched to the removal it makes: setting the feature to its corpus mean activation over all tokens (mean ablation); replacing it by its activation at the same token of another series on which it fires (resample ablation); and subtracting its decoder contribution from the model's own activation rather than from the dictionary's reconstruction (clean ablation), which leaves the dictionary's error in place. Rerunning zero ablation in the same setup reproduced the stored results exactly. Of the 91 features, 69 still clear the on-manifold null under mean ablation, 58 of 90 scorable ones under resample ablation and 49 under clean ablation; 79 clear it under at least one of the three and 37 under all three. Mean and resample ablation keep the direction of the effect (median cosine to zero ablation 0.997 and 0.991), clean ablation less so (0.919). Survival depends on how densely a feature fires: of the 62 features whose mean is below a tenth of their activation on their top series, 56 survive mean ablation, but only 13 of the 29 denser ones, for which mean ablation also adds the feature on tokens where it is silent. The alternative interventions rarely create effects where zero ablation found none: of 91 sampled non-clearing features, 1 clears under mean, 2 under resample and 8 under clean ablation (MP-10). Most causal features thus keep an effect under some on-manifold intervention, but the set of single features that clear depends on the intervention, which is a further reason to read features as a graded continuum rather than as discrete units.

Alignment with generative factors is correlational. After dead-feature revival, 154 of 10,240 TimesFM features and 115 of 6,144 Chronos-T5-Base features correlate with a ground-truth generative field beyond a permutation null; the strongest is intermittency, with $\rho = 0.881$ (MP-06). The alignment does not extend to mediation: moving the corresponding generator parameter does not change a matched feature's forecast effect beyond chance (MN-17), and some vivid candidates fail seed-stability checks (PM-11).

8.3 RQ2: Are concepts shared across architectures?

Answer. Whole-layer geometry is shared above chance by all 21 model pairs, held-out, once windows are compared without averaging. Different architectures select the same series for a concept, and this replicates held-out on both corpora and on raw real windows. The broad effect types recur in every model. At a fine grain, however, concepts are organized by model, and agreement of causal effects on the same series is detected only between two Chronos-family models.

Geometry is shared above chance but has no single meaning. With concatenated windows, peak CKA (the maximum over all layer pairs) between TimesFM and each other model is 0.358–0.635 on v2 and 0.375–0.712 on v3 (512 series each), and every one of the 21 model pairs exceeds its shuffled-series null on both corpora ($p = 0.001$, 1000 permutations; null 95th percentiles 0.003–0.51, the highest for pairs with Time-MoE). The layer pairs selected on v3 dev were registered and tested once on a fresh sealed private split: all 21 pairs exceed their nulls (Holm $p = 0.021$), with TimesFM/Chronos-2 at 0.707 [0.679, 0.734] against 0.712 on dev. Window-mean pooling understates these values, and the registered test confirms it: at the same layers and on the same private series, averaging lowers TimesFM's CKA with Chronos-2, Chronos-T5-Base, Chronos-Bolt and Sundial by 0.322, 0.433, 0.301 and 0.303 (each 95% interval above 0.26; Holm $p = 0.0008$; MN-38). Peaks between other model pairs change less, and pairs with Timer mostly fall, because Timer's 96-step token has no within-window order to match. The mean-pooled peak of the full v2 run (0.426) likewise replicates on its private split (0.411 [0.401, 0.427]) and on v3 (SH-01). Different similarity metrics, however, rank the 21 model pairs inconsistently (Kendall's $W = 0.25$, $p = 0.0095$, over the seven metrics of the full v2 run, whose CKA is mean-pooled; Figure 8). TimesFM/Chronos-2 is the most similar pair by error agreement (0.948) but only 15th of 21 by mean-pooled CKA, where Sundial/Chronos-Bolt ranks first (0.927). Part of this gap comes from pooling: with concatenated windows TimesFM/Chronos-2 ranks 9th on v2 and 7th on v3, while Sundial/Chronos-Bolt stays first ($\rho = 0.82$ and 0.76 between the two rankings on the same series). The report therefore never pools similarity metrics into one score. Two finer-grained measures fail their controls on real activations: an untrained pair scores higher than a trained, related pair (MN-01), and a measure of unit-level correspondence rises when both bases are rotated (SH-19). CKA is accordingly read only against shuffled-series nulls and never as evidence of lineage.

Models select the same inputs for a concept. On dev, 1,296 of 1,584 atlas transfer tests are reciprocal after Benjamini–Hochberg correction. Twenty transfer claims were registered from dev results alone, all derived from two source concepts: TimesFM layer 18 (14 claims) and Sundial layer 6 (6 claims). All 20 confirm on the 971 private series, each at the smallest Holm-adjusted p attainable for 20 claims at 2,000 null draws (0.0100), with AUCs of 0.919–0.970 against null 95th percentiles of 0.70–0.75 (Figure 6b). The destinations span all seven models. Only 8 of the 20 best-matching private features are the ones selected on dev, so the confirmed claim is that the destination layer contains *some* feature selecting the same series, not that a particular feature pair corresponds. On v3, a new set of 20 claims, from Sundial (14), Chronos-2 (5) and Time-MoE (1), confirms 20 of 20 on the private split and 20 of 20 on the 800 raw GIFT-Eval windows (AUC 0.757–0.991). The same transfer protocol was confirmed in three earlier one-shot looks with their own claim sets (19 of 20,

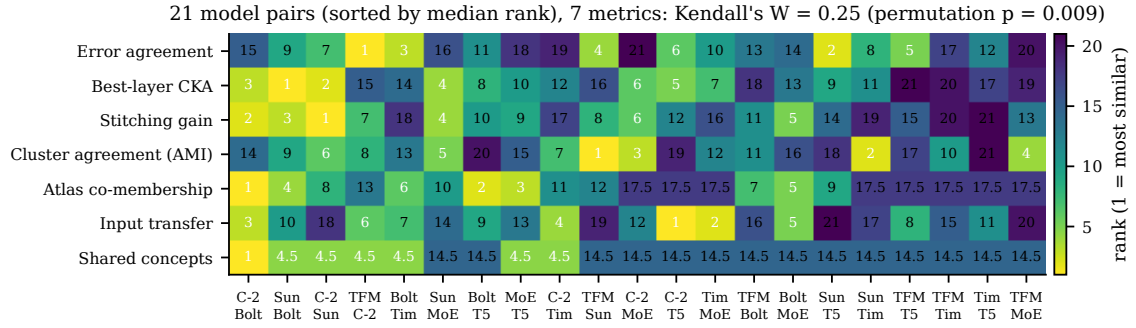


Figure 8: Rank of each model pair (columns, sorted by median rank; TFM TimesFM, C-2 Chronos-2, Sun Sundial, Bolt Chronos-Bolt, Tim Timer, MoE Time-MoE, T5 Chronos-T5-Base) under seven similarity metrics (rows); 1 is most similar. Source: `report/model_similarity.json`.

then 20 of 20 twice; SH-18). The result is also insensitive to the alignment window: re-pooled at 64 and 96 steps, the v3 reciprocal rate moves from 0.827 to 0.801 and 0.799, and 92% of test verdicts are unchanged (SH-23). This is evidence about inputs, not effects.

The broad effect types recur in every model. The three largest families, level raisers, level lowerers and seasonality dampeners, each have causal members in all seven models, and volatility dampeners in six (Figure 7b). Membership reflects each feature's own measured effect, so it does not depend on whether the family partition beats its shuffle null. The observation is descriptive and exploratory; Timer's members come from a model whose clear rate is at chance.

Fine-grained concepts are organized by model. Of the 11 atlas concepts, 9 have causal members from at least two models, but none spans more than 3 of the 7 models. The clusters are also *purer* by model than random assignment would make them: there are 9 multi-model concepts against a null mean of 10.8 ($p_{\text{below}} = 0.025$), and the mean model purity is 0.591 against 0.474 (SH-22). By sharing class, 2 concepts are shared, 2 partially shared, 5 convergent (the same effect reached on different series) and 2 single-model (Figure 9a); among the 9 seed-stable concepts the counts are 2, 1, 4 and 2. The structural claim holds held-out on both corpora: no concept of the private atlas spans more than 3 of 7 models on v2 (10 concepts) or more than 2 on v3. At the grain of the atlas, each architecture thus builds its own variants of an effect, and several arrive at the same effect independently.

The same effect on the same series is detected only within the Chronos family. The shared-input test ablates both sides of a reciprocal transfer on the same series and requires each side to clear its own null. Of 1,290 tests (at most 60 per ordered model pair), 216 are scorable (Figure 9b). Among them, 7 show the same causal effect, 54 agree on level or shape alone, 133 show no specific agreement and 22 act differently. All 7 "same" verdicts are between Chronos-2 and Chronos-Bolt, six of them through one feature set in Chronos-Bolt's fourth block. Of the 1,074 unscorable tests, 1,053 fail because one side's own ablation does not clear its null on the 21–40 shared series (median 32). Because the test is

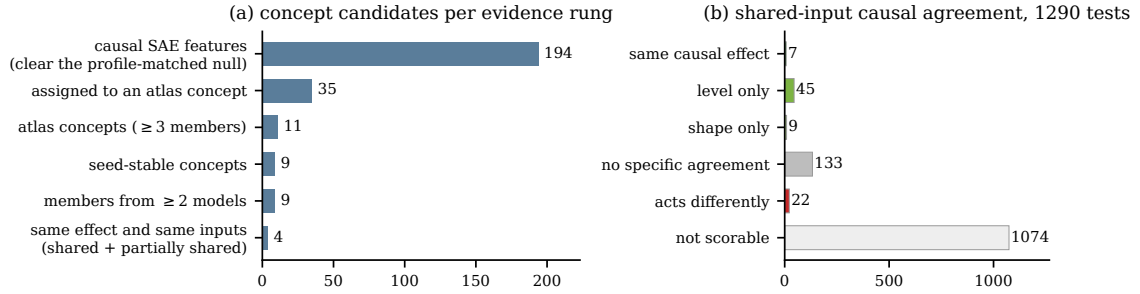


Figure 9: (a) Concept candidates at each rung (the first three bars are sequential; the last three are properties of the 11 atlas concepts). (b) Shared-input causal agreement over 1,290 tests drawn from the 1,296 reciprocal transfers. Source: `sae/concept_stage.json`.

specific but insensitive (Section 6), the 7 detections stand, while their absence elsewhere is not evidence that effects differ.

Cross-model causal claims do not confirm. None of the claims that two models share a causal effect replicates held-out. On v2, the three registered multi-model atlas concepts fail the registered conjunctive rule. Their private centroid cosines (0.955–0.988) exceed the null's 95th percentile (0.854–0.907; $p \leq 0.007$), but the rule also required half of the member pairs to reach cosine 0.9, and with three members only one pair did; reading the centroid component alone would be post hoc. Of the registered shared-input agreement claims, 0 of 6 tested confirm and 5 of 11 are not testable; the best, between Chronos-Bolt block 4 and Chronos-2 block 7, reproduces its verdict ($p = 0.011$) but misses the Holm threshold (0.066). On v3 no cross-model causal claim could be registered, because all four atlas concepts sat on targets that failed the per-target significance gate set before the data were generated ($q \geq 0.243$).

8.4 RQ3: Do concepts causally affect the forecast?

Answer. Single features do causally shape the forecast, and some of these effects replicate on new series at the attainable significance floor. The effects are small, fewer than half replicate, and neither activation prominence nor attention mass identifies them; a causal test is required.

Single-feature effects replicate partially. Of the 32 single-feature causal claims registered from v2 dev, 31 were testable on the private split and 10 confirm, each with $p = 0.001$ after an adaptive redraw to 1,000 null draws and $p_{\text{Holm}} = 0.031$, the floor for 31 tests. By model, the confirmed share is highest for TimesFM (3 of 4) and lowest for Chronos-2 (1 of 7) and Chronos-T5-Base (0 of 4); Time-MoE, Sundial and Chronos-Bolt confirm 2 of 7, 2 of 4 and 2 of 6. By channel, near-horizon shape (4 of 8), MASE (3 of 8) and spectral centroid (2 of 4) confirm most often, and the level and seasonal-strength claims (4 each) never do. Of the 21 claims that are not detected, 19 had not reached the 200-draw p floor on dev, and one changes sign (Chronos-T5-Base block 10). On v3, under the stricter null, 11 of 24 testable claims confirm.

Activation prominence does not identify causal importance. Of the 927 most prominent features of the private atlas, 0.835 have no causal effect (one-sided lower bound 0.817; dev 0.810; $p_{\text{Holm}} = 0.0005$; MN-14); on v3 the share is 0.919 of 861. TSFM-Lens therefore ranks features by measured effect rather than by activation.

The effects are small. The median single-feature effect is 0.097 context standard deviations, 2.2 times smaller than the 0.214 by which substituting the SAE reconstruction moves the forecast (v3: 0.156 against 0.260, 1.7 times; MN-15). Among the cells that clear in the v2 run, the median effect is 1.24 times the null's 95th percentile.

Attention mass is not causal importance. The attention mass that a head places on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ in TimesFM and -0.118 in Chronos-T5-Base; CA-04), consistent with Jain and Wallace (2019). A minimal seasonality circuit is one head in TimesFM but five non-additive heads in Chronos-T5-Base (CA-03).

8.5 What limits single-feature causal evidence

The v3 run tested three ways of strengthening single-feature evidence, each judged by a criterion set in advance. None made individual features more specific than on-manifold random directions.

- **More training data for the dictionaries.** Three times as many training rows raised the median held-out fidelity from 0.749 to 0.82. Across targets, however, fidelity is, if anything, negatively correlated with the share of tested features that clear the null ($\rho = -0.326$, $p = 0.022$, $n = 49$; MN-34).
- **A more powerful per-feature test.** Using 32 series and 64 null draws per feature, with multiplicity control, fails the planted-concept criterion that the 8-series rule passes (Table 3; MN-36).
- **Choosing candidates by gradient attribution.** Ranking atoms by |activation $\times$ gradient of each horizon step| first passed its own planted test. It nominated 620 additional features and found more causal ones (112 instead of 78 on 38 targets), but at no higher rate than random candidates (0.085 against 0.110, $p = 0.335$), although it does beat random candidates against the isotropic null (0.299 against 0.192, $p = 0.002$; MN-37).

The binding limit is therefore the size of a feature's effect relative to on-manifold directions, not the amount of data, the dictionary's quality or the choice of features to test. Since fidelity does not predict causal clarity, the dictionary admission gate now uses fidelity only as a sanity floor of 0.5, with forecast preservation measured on 128 series; under this gate 35 of 50 v3 dictionaries are admitted instead of 8, without retraining (MN-34).

A related question is whether the forecast change that a dictionary can express is carried by a few atoms. On 31 targets, zeroing the five attribution-ranked atoms of a layer produces a median share of 0.74 of the combined change from the features and the dictionary error, and on 30 of the 31 targets this exceeds activation-matched random sets of five atoms (descriptive; D-42C). The expressible forecast change is thus concentrated in a few atoms,

but those atoms mostly do not act more specifically than a typical on-manifold direction of the same size.

8.6 RQ4: Where in depth, and how comparable is depth?

Answer. The way TimesFM and Chronos-2 react to corruptions across depth replicates held-out for 6 of 9 corruptions on v2 and 4 of 9 on v3, with 8 of 9 keeping their sign; one corruption, level shift, is unstable across corpora. Absolute depth locations are comparable only after accounting for captured compute and the depth convention.

Depth-resolved reaction profiles replicate. For TimesFM and Chronos-2, the rank correlation between their per-layer corruption fingerprints replicates on the v2 private split for 6 of 9 corruptions (overall $\rho = 0.524$ [0.504, 0.544]; Figure 6a). The two models react at matching depths to detrending (private $\rho = 0.368$) and dropout (0.510), at opposite depths to noise (-0.743), spikes (-0.785) and warps (-0.651), and not systematically to deseasonalizing (0.086 [-0.027, 0.186]). Under the registered rule, which requires the dev value to fall inside the private interval, three corruptions do not replicate. Two of them show a *stronger* effect on private data: frequency shift (0.381 against 0.216 on dev) and smoothing (-0.581 against -0.418). The third, level shift, changes sign (dev -0.188; private 0.109 [-0.025, 0.159]). On v3, 4 of 9 corruptions replicate under the strict rule and 8 of 9 keep their dev sign. Level shift flips sign again (dev -0.164; private 0.252), so it is the one depth agreement that is not stable across corpora.

Where restoration happens. In TimesFM, restoration by per-window patching concentrates in the window just before the horizon and grows with depth, from 0.427 at layer 0 to 0.612 at layer 8 and 0.631 at layer 16 for the noise corruption, while restoration in the first window falls from -0.191 to 0.000 (320 v2 series; DE-01).

Depth locations need two qualifications. Captured compute ranges from 13.9% (Chronos-T5-Base) to 99.8% (Timer) of a forecast's FLOPs (DE-05), and Chronos-T5's last encoder block sits at depth 0.478 on the whole-stack axis (DE-04); the report renders both next to every depth claim. Moreover, no layer chosen by one of three layer selectors was robust to the choice of selector for any model (DE-08).

8.7 Can internals predict forecast failure?

If internal features carry causal information, they might tell a user when to distrust a forecast. For each model we predict a series' log MASE (scored by Spearman correlation) and a failure flag (scored by AUROC) from free, output-only features: quantile width, flatness and catch22 features of the context. We then add the SAE-family activations at the last window and measure the gain, cross-fitted with a series bootstrap on 941 dev series ($n_{\text{boot}} = 1000$). The evidence class is PREDICTIVE: a feature that predicts error is not shown to cause it.

The free baseline is strong (Spearman 0.76–0.85). For five of the seven models (Chronos-2, TimesFM, Sundial, Chronos-T5-Base and Chronos-Bolt), the gain's interval includes zero for both tasks (log-MASE Spearman gains -0.004 to $+0.005$, AUROC gains -0.010 to $+0.008$). The gain is positive for the two weakest models, Timer ($+0.030$ [0.019, 0.043]; baseline 0.787; failure rate 0.30) and Time-MoE ($+0.012$ [0.005, 0.019]; baseline 0.853; failure

rate 0.62), robustly across five fold seeds; no failure-AUROC gain excludes zero. Both gains were registered and confirm when the frozen procedure is refit on the private split ($n = 935$): Timer 0.0446 [0.0292, 0.0605], $p_{\text{Holm}} = 0.002$, and Time-MoE 0.011 [0.0032, 0.0194], $p_{\text{Holm}} = 0.012$. Fourteen dev comparisons were run and two were selected, which is legitimate because the private split referees the selection. On v3 both confirm again, while a newly registered Chronos-T5-Base claim does not.

Internals do not help choose a model. Routing each series to the model with the lowest predicted MASE gives a mean MASE of 1.948 [1.814, 2.070] with the baseline features and 1.964 [1.837, 2.093] with internals added (difference -0.016 [-0.044, 0.010], $p = 0.258$), and both are worse than always using Chronos-2 (1.856; $p = 0.001$; oracle 1.523). Internals add information only for the two least accurate models, whose forecasts fail most often.

RQ	Result (v2 unless stated)	Evidence class	Status	ID
1	442 feature-channel cells clear an isotropic null, 2.47× empirical chance; Timer 1.02×	causal within-model	exploratory	MP-07, PM-17
1	Against an on-manifold null, features exceed chance only modestly (v3: 91 vs. 70.4, 1.29×)	causal within-model	exploratory	CA-12
1	9 families cover 99% of features but do not beat a shuffle null ($p = 0.68$)	descriptive	exploratory	CA-10
2	All 21 pairs above null with concatenated windows; TimesFM/Chronos-2 0.712 dev, 0.707 fresh private; the window mean understates it by 0.30–0.43	geometric	held-out confirmed	SH-01, MN-38
2	7 similarity metrics agree only weakly ($W = 0.25$)	mixed	exploratory	SH-05/06
2	Untrained pair more similar than related trained pair	geometric	confirmed negative	MN-01
2	Concept transfer on inputs: 20/20 on v2 and v3, 20/20 on raw real windows	correlational	held-out confirmed (2 corpora)	SH-18
2	Level and seasonality-damping causal features in all 7 models	descriptive	exploratory	CA-10
2	No concept spans more than 3 of 7 models	descriptive	held-out confirmed (2 corpora)	SH-16
2	Causal atlas purer by model than chance ($p_{\text{below}} = 0.025$)	descriptive	exploratory	SH-22
2	Same effect on same inputs: 7/216 scorable, all Chronos-2/Bolt; test specific (0/9 decoys)	causal on shared inputs	exploratory	SH-22, MN-30
2	Multi-model atlas concepts 0/3; shared-input agreement 0/6	causal, compared	not confirmed	CA-11
3	Single-feature causal effects: 10/31; v3 11/24	causal within-model	partly confirmed	CA-11/12
3	83.5% of prominent features causally null (927 held-out; v3 91.9%)	causal within-model	held-out confirmed	MN-14
3	Feature effect 2.2× below SAE reconstruction cost (v3 1.7×)	causal within-model	exploratory	MN-15
3	Attention mass uncorrelated with head causality	causal vs. descriptive	exploratory	CA-04
3	More data, a powered test or attribution candidates do not raise specificity	causal within-model	negative	MN-34/36/37
4	Depth-profile agreement 6/9 corruptions; v3 4/9; level shift flips on both	causal within-model	partly confirmed	CA-01
4	Captured compute 13.9%–99.8% qualifies depth claims	descriptive	–	DE-05
–	Internals predict failure for the two weakest models only	predictive	held-out confirmed (2 corpora)	MN-28
–	Chronos-2 more accurate than TimesFM on `mixture` (0.209; v3 0.303)	behavioral	held-out confirmed (2 corpora)	PM-05

Table 8: Headline results with evidence class and status. Each ID is an entry in `FINDINGS.md` with exact numbers, reproduction command and artifact path.

9 Discussion

Shared inputs, model-specific mechanisms. The most robust cross-model result is about inputs: models with different tokenizations, depths and training objectives single out the same series for a concept, on two corpora and on raw real windows. Whole-layer geometry is shared as well: all 21 model pairs exceed shuffled-series nulls on both corpora and on a fresh sealed split, most strongly at early, input-encoding layers, once windows are compared without averaging away their internal order. Yet at the grain of individual causal effects, concepts are organized by model, and agreement on the same series is detected only between two closely related Chronos models. One consistent reading is that the panel converges on *what* in a series matters (its level, its seasonality) while implementing the response differently. This is a weaker form of convergence than the representational convergence argued for in vision and language (Li et al., 2016; Huh et al., 2024), and our instruments locate the boundary: input agreement is measurable and replicates, whereas effect agreement across architecture families is below what ablation can resolve at these effect sizes (Section 6).

What the causal evidence supports. Single SAE features clearly beat isotropic random directions of the same size but barely beat directions drawn from the activation distribution. More dictionary data, a more powerful test and gradient-selected candidates do not change this (Section 8.5). We read this as a property of the models at this level of description: the forecast-relevant information in a layer is spread over many directions, and removing any one typical direction moves the forecast by a comparable amount. Claims that a single feature "is" a forecasting concept should therefore be stated relative to an on-manifold null, since against an off-manifold null a typical direction already looks special; this is the time-series counterpart of the illusions documented for subspace interventions (Makelov et al., 2024).

Implications for method. Three lessons generalize beyond this study. First, activation-based selection is not a substitute for intervention: most prominent features have no causal effect, and attention mass does not predict head importance. Second, nulls must be validated on a known answer before they are trusted; the plausible uniform null inflated the count of causal features 2.7-fold (293 against 110) and failed the planted test on every seed. Third, a negative verdict needs a test of sensitivity: our agreement rung never produces a false "same", but it also rarely detects a planted shared concept, so its silence carries no information. Exploratory pipelines with thousands of comparisons (Gelman and Loken, 2014) benefit from the same register-then-confirm discipline used here, which in our study confirmed between half and two thirds of the registered claims (43 of 82 and 41 of 63).

Practical use. The internal features add information about forecast failure only for the two least accurate models, whose forecasts fail most often, and do not improve model selection. For practitioners, the more immediate value of the system is diagnostic: behavioral checks such as calibration and missing-value handling, and the documented failures of instruments that would otherwise produce confident, wrong results.

10 Limitations

Models. The study covers seven transformer models with contiguous tokens; models whose tokens read disjoint lags, such as Lag-Llama, receive reduced scope. With seven models,

contrasts between models are confounded with architecture, size and training data. Timer's features do not exceed chance, so its results carry no evidence about causal concepts. The analysis is univariate: Chronos-2's cross-series group attention is not measured (PM-13), and Chronos-T5's decoder and Sundial's flow head are not captured.

Time alignment. A common time grid is no finer than the coarsest token: no window rule recovers resolution finer than Timer's 96-step token, and peak layer locations depend on the window size (SH-23). Concatenated windows, which the similarity results use, are computed on 512 series per corpus, the size at which the Gram matrices fit in memory. SAE comparisons do not depend on the window rule. Dictionaries are trained on window averages, which is conservative for 1-step models: trained on its own tokens, Chronos-T5-Base's dictionary explains token variance far better and yields more causal features, while the top series it shares with TimesFM's dictionary change within seed noise (MP-11).

Ground truth and external validity. Ground truth is exact only for the synthetic tier and encodes the generators' assumptions; 12 of 33 concept parts are dominated by one real-derived generator, and overlap with the models' pretraining data cannot be verified. Real data are part of the analysis: 410 of the 965 v2 dev series are real-derived, and the dictionaries are trained with about 4,000 additional real Monash windows per target. Raw real benchmarks are nonetheless a weaker basis for the causal questions. Real rows give TimesFM representations of lower effective dimension than the synthetic corpus (1.76 against 2.66) and more dead dictionary atoms (MN-20); weakly predictable real-derived series receive flat forecasts in 67–100% of cases, leaving an ablation nothing to move (BM-06); and public archives may be in the models' pretraining data. Raw real data therefore enter as an external check rather than as a discovery corpus: concepts are discovered where ground truth exists, and the v3 transfer claims are then confirmed on 800 raw real windows held out from every dev stage (20 of 20).

Scope of the confirmation. Held-out confirmation covers accuracy, geometry, perturbation agreement, transfer, single-feature causal effects, atlas structure and failure prediction, on two corpora with one private split each, and the registered window-geometry claims were tested on a second sealed v3 split. The 20 confirmed v2 transfers derive from two source concepts and the v3 transfers from three source models, so the transfer result rests on a small number of concepts. Registered claims are selected from larger dev pools; this is legitimate because the private split referees them, but it means that unregistered dev results remain exploratory.

Effect sharing on the same series. Per-series agreement of causal effects across architectures is not measurable by ablation at the effect sizes these models have. Both the shared-input rung and its subspace redesign fail the sensitivity requirement of the known-answer test (Section 6), and TSFM-Lens therefore compares effects across models at the level of concept profiles.

Dictionaries, interventions and usability. Concept results depend on dictionary size, sparsity, revival and seed; the system measures this dependence but cannot remove it, and feature death tracks layer geometry (alive atoms scale with effective dimension with a log-log exponent of 1.02, $\rho = 0.58$ over 45 cells; MP-05). Which single features clear the null

depends on the intervention: of 91 v3 causal features, 37 clear it under mean, resample and clean-activation ablation alike (MP-10). The controls catch only failures for which a known-answer case exists. Readability for newcomers is a design goal; no user study has measured comprehension.

11 Conclusion

Interpreting TSFMs is harder than it looks: tokens mean different things in different models, real series have no annotated ground truth, and causal effects are small enough that measurement artifacts dominate unless they are controlled. TSFM-Lens addresses these problems with a controlled benchmark, measured time alignment, null- and reach-gated causal concepts validated on a planted forecaster, and a register-then-confirm protocol, behind one command and a report written for newcomers. Across seven TSFMs and two corpora, different architectures share above-null representational geometry, a similarity that the common window average had hidden, and select the same inputs for a concept, while their causal machinery is organized by model; causal features act mainly on level and seasonality but are only weakly specific against on-manifold directions; activation prominence does not reveal causal importance; and depth-resolved reaction profiles replicate for most corruptions on the primary corpus and about half on the second. Each conclusion is stated with its evidence class, and each held-out claim was registered before the private data were opened.

Availability. Code (MIT license), configurations, the example run and the findings ledger are at `https://github.com/DanK10097110/TSFM-Interp`. Version 1.1.0, which this paper describes, is archived at doi:10.5281/zenodo.23267462, and both packages install from PyPI (`pip install tsfm-lens tsfm-benchmark`).

Acknowledgments and Disclosure of Funding

Computational resources were provided by the second author's lab at Rutgers University. This work received no external funding, and the authors declare no competing interests. Use of AI tools: Claude (Anthropic) wrote most of the repository's code and tests, ran the experiments and helped draft and edit the manuscript, under the authors' direction. The research questions, ideas and general design choices are the authors', who reviewed the implementation details Claude proposed, checked every reported result and take full responsibility for the content.

Appendix A. Implementation and Usage

Repository layout. `tsfm_benchmark` has one subpackage for generation, audit and sealing and another for diversity and exchangeability validation. `tsfm_lens` has subpackages for models, extraction, analysis, sparse autoencoders and the report, plus the stage orchestrator `pipeline.py`. The folder `examples/panel7_v2` contains the report, configuration and every lightweight artifact of the v2 run, so the report and this paper's figures can be regenerated without a GPU.

A typical session. One installs the two packages, builds a benchmark and runs a configuration:

```
pip install -e . && pip install -e tsfm_lens
cd tsfm_benchmark && PYTHONPATH=. python3 example_runs/run_full.py \
       --config configs/example.yaml --out ./benchmark_out --references monash
cd ../tsfm_lens && python run.py --config configs/examples/medium_2model.yaml
```

The command-line interface adds a preflight check (`--doctor`), partial reruns (`--stages`, `--force`), checks for new checkpoints or library upgrades (`--check-alignment`, `--check-adapter`) and an environment diff (`--verify-provenance`). A CPU-only mock configuration runs every stage in minutes in continuous integration, and a worked example (`tsfm_lens/docs/worked_example.md`) walks through a report. In the v2 run, stages up to and including L3 take about 1.4 hours; most of the 21.1 hours are spent on SAE training and the concept battery.

Adding a model. A new checkpoint first tries the zero-code adapter. Of eight TSFM checkpoints probed, two (Timer and Time-MoE) load this way, and five that ship as standalone packages need a hand-written adapter (MN-25). For these, the repository provides a template (`tsfm_lens/tsfm_lens/models/TEMPLATE_adapter.py`), a conformance check (`--check-adapter`, which runs patch-identity and reach checks, the alignment gate, span discovery and capability checks) and a guide with a checklist of pitfalls (`tsfm_lens/docs/ADDING_A_MODEL.md`).

Extending the analysis. New report sections declare the artifacts they require, new claims must use the typed `Finding` record, and new configuration fields declare which stage artifacts they affect, so that the fingerprints mark exactly those as stale. New generator options are opt-in, so that sealed corpora still regenerate bit-exactly.

The example report. The seven-model report (`examples/panel7_v2/report.html`) contains 530 "What does this mean?" notes, one for every figure, and 240 typed findings (89 descriptive, 54 behavioral, 49 causal, 42 linear-translatable, 4 geometric and 2 illustrative). A glossary and a methods appendix are generated from the code. The complete analysis of the four-model preset is one command, `python run.py --config configs/examples/large_4model.yaml`, and smaller presets cover a CPU-only smoke test, a single model and a pair.

Appendix B. Additional Validity Analyses

This appendix collects the evidence on threats to validity that a careful reader is likely to raise, grouped by topic. Each item gives the evidence, its source and what remains unmeasured. Numbers are from the v2 run unless stated otherwise.

Benchmark quality and split exchangeability. Every sample passed a banded-DTW leakage gate (threshold 0.35) against Monash reference series: 68 of 2,000 candidates were rejected, and the closest accepted sample sits at distance 0.352. The dev split and the private epoch used for confirmation are exchangeable: energy-distance $p = 0.951$, 20 of 22 catch22 features equivalent by TOST (the other two inconclusive at this n), tier and family composition indistinguishable ($p = 0.94$ and $p = 0.9999$), and 6 near-duplicate pairs among

937,015 cross-split pairs. Diversity is measured in catch22 space: the dev split has effective dimension 5.37 of 22 features (private 5.57) and a near-collision fraction of 0.052. Each sample is labeled with its tier, so every result can be split by tier. The v3 corpus passes the same checks: its dev split (2,876 series) and private epoch (2,868 sealed series, of which `confirm` draws 1,024) give energy-distance $p = 0.995$, all 22 catch22 features equivalent by TOST, tier composition $p = 0.97$, and 9 near-duplicate pairs among 8,248,368 cross-split pairs. This check is model-free and was computed after the confirmation look. The realism report compares raw moments and is not informative across scales; its only usable statistic, spectral centroid, differs from the reference by 0.022.

Why a synthetic benchmark rather than raw real data. There are four reasons, each measured. Ground truth: only generated series record their components, so only they can validate a probe (P1). Representation: on real Monash rows TimesFM's representations have lower effective dimension than on the synthetic corpus (1.76 against 2.66) and more dead dictionary atoms (MN-20). Measurability: weakly predictable real-derived series receive flat forecasts in 67–100% of cases (BM-06); across ablation panels, noise-like contexts are flat 82.3% of the time against 8.5% for smooth ones, and a flat forecast gives an ablation nothing to move. Contamination: public archives may be in the models' pretraining data. Real data are still used, each for a stated purpose: real-derived series (410 of 965) for realism and per-tier checks, about 4,000 raw Monash windows per target only as extra dictionary-training rows, and the Monash archive only as the leakage reference. The v3 corpus labels each series with one of these roles and adds a fifth: 800 raw GIFT-Eval windows held out from every dev stage, used only to re-test the registered transfer claims (20 of 20 confirm).

Statistical power and data sufficiency. Sufficiency depends on the claim, and each claim's power is stated. Not power-limited: the 20 transfer claims all reach the Holm floor (AUC 0.919–0.970 against null 95th percentiles of 0.70–0.75); the accuracy claim's minimum detectable ΔMASE is 0.133 against an observed 0.209; the held-out concatenated CKA of TimesFM and Chronos-2 is 0.707 $[0.679, 0.734]$ against a null 95th percentile of 0.010. Power-limited: single-feature causal claims (each feature is tested on its 8 strongest-firing series, and 19 of the 21 that did not confirm never reached the dev p floor), per-target concepts (178 of 672 candidates never fire on any series, so more rows do not help; SH-20), and the shared-input agreement rung, which more shared series do not rescue (MN-30). More data was then tested directly. A three times larger corpus (v3, 2,876 dev series) reproduces the held-out pattern (41 of 63 claims) but does not make features more causal. Neither does a per-feature test with four times the series and null draws, which is less sensitive on the planted forecaster (Section 8.5; MN-34, MN-36).

Why concepts are defined by causal effect. Correlation measurably misleads here. 83.5% of the most prominent features have no causal effect on held-out data (MN-14); attention mass on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated (CA-04); two sparse features can correlate at $\rho = 0.80$ with no shared top series (MN-19); features aligned with a ground-truth field do not mediate that field's effect (MN-17); and on a planted forecaster a decoy concept correlates with forecast error in five of five seeds while its causal effect is null (MN-31). An activation-based concept can be real and still irrelevant to the forecast; the causal definition asks the question that matters for forecasting.

Fairness of zero ablation. For a TopK dictionary, zero is a feature's resting value: at most 32 of 192–6,144 atoms are nonzero on any token, so the ablated state lies inside the dictionary's normal range, unlike zeroing a dense neuron. The comparison is between two passes through the same reconstruction, so the dictionary's error is in both arms. The null removes exactly the feature's per-token amount along another direction, on the same series, with the same seed for sampled models, after a reach probe has shown that patching this layer can move the forecast. Each alternative intervention, with the null matched to its removal, keeps most of the 91 v3 features that clear the on-manifold null (Section 8.2): mean ablation 69, resample ablation 58 of 90 and ablation against the clean activation 49, with median effect-vector cosines to zero ablation of 0.997, 0.991 and 0.919. To make this comparison possible the v3 dictionaries were retrained from the recorded configuration; training reproduced all 50 dictionaries exactly (identical loss histories) and the same 91 features clear.

Distinguishing causal features from chance. Four checks address this. (i) Calibration on a known answer: on the planted forecaster's control layer, which has no planted effect, the per-cell false-positive rate is at most 0.003 while planted effects are found with sensitivity 0.70–1.00 (MN-29). (ii) Counting against an empirical chance rate, estimated per cell by scoring each null draw against the others (median 0.018): 442 feature-channel cells clear against 178.75 expected (2.47 times chance), and 22 of 55 targets are individually significant (BH). At the level of whole features the excess is smaller: 194 of 1,022 scorable candidates clear at least one channel against at most 144.95 expected if channels were independent. (iii) Held-out replication: 10 registered single-feature claims confirm on new series at the attainable p floor after adaptive redraws to 1,000 null draws. (iv) Null strictness: a stricter null that draws the random direction from the layer's own activation covariance also passes the planted test (sensitivity 0.916). On v3, where both nulls were scored on the same features, clearing cells fall from 4.92 times chance under the isotropic null (600 against 122.1 expected) to 1.27 times under the covariance-matched null (170 against 133.5). Causal effects therefore exceed random directions clearly and typical on-manifold directions only slightly. The v3 run registered every causal claim against the stricter null. There, 91 of 967 features clear at least one channel against 70.4 expected when channel dependence is kept (1.29 times, $p = 0.0045$; Table 7), and 11 of 24 registered claims replicate on new series (CA-12). The stricter null leaves few causal features, and about half of the registered ones replicate.

Effect sizes. Effects are small relative to the forecast and to the dictionary's own error. The median single-feature effect is 0.097 context standard deviations, 2.2 times below the change from substituting the reconstruction (v3: 1.7 times; MN-15); among the cells that clear in the seven-model run, the median effect is 1.24 times the null's 95th percentile. The paper therefore makes no claim that single features explain forecasts; it claims that specific, replicable directions exist and characterizes what they do.

Robustness of the sparse dictionaries. Each target has three replicate dictionaries, and 9 of 11 atlas concepts are stable across them. Held-out fraction of variance explained has medians of 0.647 (TimesFM) to 0.878 (Chronos-T5-Base) across models, and only 5 of 56 targets pass the strict admission gate (held-out fidelity ≥ 0.85, forecast change under substitution ≤ 0.25 MASE, dead-rate bound). With three times the training rows (v3), the

median held-out fidelity rises to 0.82, but fidelity does not predict causal clarity ($\rho = -0.326$ across 49 targets; MN-34). The ablation keeps the dictionary's error in both arms, so a measured effect is a real intervention whatever the fidelity. Fidelity is therefore now only a 0.5 sanity floor, and forecast preservation is measured on 128 series; under this gate 35 of 50 v3 dictionaries are admitted. Joint crosscoders were tested as the alternative and lost to independent dictionaries matched afterwards (0.2787 against 0.3204, $p = 0.002$; SH-10), which is why TSFM-Lens uses one TopK dictionary per layer.

Details of time alignment. Alignment has three steps, all measured rather than declared. (1) Each token's span is measured by injecting an impulse at every time step and recording which token responds most, gated on peak-to-pedestal contrast (≥ 4) and contiguity (≥ 0.95); measured spans equal the hand-written adapters' on six checkpoints (IoU 1.0) and recover Timer's 96-step tokens (MP-01). (2) Tokens are overlap-pooled onto common 32-step windows as described in Section 3.4. The context of 480 steps is a multiple of 96 so that Timer's tokens tile it exactly (at 512 its span discovery refuses). (3) An alignment gate injects an impulse into each window and checks per layer that the matching window moves most; all models pass at their shallowest layer, and the gate's per-layer record shows that deep layers of Sundial, Time-MoE and Timer lose time localization (diagonal hit 0.07–0.27 against chance 0.067), which the report renders and which qualifies window-level claims there. Values are normalized by each model's own input scaling, verified by a scale-equivariance test (residuals 2.7×10^{-7} to 0.037 context standard deviations); effects are compared across models in units of each feature's own null 95th percentile; depth is the fraction of the whole stack. Re-pooled at 64 and 96 steps (v3), the transfer rate moves from 0.827 to 0.801 and 0.799, with 92% of verdicts unchanged, and peak CKA by a median of 0.013 and 0.018. The peak *layer pair* is the same at all three windows for only 9 of 21 model pairs, so peak locations are window-dependent (SH-23).

Sampled models. Every compared pair of calls to Sundial or Chronos-T5 uses the same seed. Unseeded, back-to-back Sundial calls differ by up to 0.50, and a null seeded differently from its baseline once made Sundial look causally inert (PM-09); repeat-run noise floors are reported per model (Sundial 0.14, Chronos-T5-Base 0.24 mean $|\Delta\text{MASE}|$, the deterministic models 0).

Integrity of the confirmation. The registry (128 hypotheses, sha256 recorded) was frozen before the private epoch was generated; the epoch shares no sample hash with any of seven earlier splits (one candidate epoch was discarded unopened for overlapping an old split); `confirm` ran once, and a crash before any statistic was written is disclosed in Appendix D. Registered claims are cuts from larger dev pools (transfer 20 of 1,124 candidates, causal 32 of 113, agreement 11 of 25, atlas 4 of the 9 stable concepts), which is legitimate because the private split referees them, and is why only registered claims are called confirmed. The v3 registry (111 hypotheses, sha256 `67b91269...`) was frozen in the same way before its private epoch was minted; its external windows share no sample hash with the dev or private split.

Multiple comparisons. Dev results are exploratory and labeled so. The ledger groups 4,238 tests into 123 named families with Holm or BH and flags families that cannot reach

significance; the battery's 8,705 cells are judged by their count against the empirical chance rate. Held-out claims use Holm per registered claim type.

Use of CKA. Given its known problems, CKA is used only against a shuffled-series null at the same layer pair with a series-level bootstrap, reported as one of seven pair metrics that are never pooled, and never read as lineage, since an untrained pair can score higher than a trained one (MN-01). The reported peak is a maximum over layer pairs; its held-out replications (0.707 concatenated on a fresh split, 0.411 window mean) are measured at the pairs selected on dev.

Choice of settings. Battery settings ($k = 8$ rows, 16 null draws) were fixed on a four-model development run before the seven-model concept stage ran, after larger settings did not recover power (SH-20); at the atlas threshold of 0.9 the v2 atlas has 11 concepts against a shuffle-null mean of 3.38 (95th percentile 5.05, $p = 0.005$); family silhouettes are flat (0.25–0.32) across cuts from 6 to 16, which is why families are reported as a reading vocabulary.

Generalization to other models. The pipeline is model-agnostic behind the adapter interface, and two of the seven models load through the zero-code path. Of eight checkpoints probed, five crashed in their configuration loader and need hand-written adapters (MN-25); models whose tokens read disjoint lags (Lag-Llama) receive behavioral analysis only.

Appendix C. Statistical Details

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}}$ resampled means $\bar{d}_b^*$, the interval is the percentile interval and $p = \min(1, \max(\frac{1}{B}, 2\min\{\frac{1}{B}\#(\bar{d}_b^* \le 0), \frac{1}{B}\#(\bar{d}_b^* \ge 0)\}))$ (`analysis/stats.py`; floor per Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Holm: $\tilde{p}_{(i)} = \max_{j \le i} \min(1, (m - j + 1)p_{(j)})$ (Holm, 1979); with $p \ge 1/B$ the smallest attainable value is m/B, which the ledger reports per family, flagging $m/B > \alpha$. BH: $\tilde{q}_{(i)} = \min_{j \ge i} \min(1, \frac{m}{j} p_{(j)})$ (Benjamini and Hochberg, 1995).

Profile-matched ablation null. For feature i with rows R_i and $J = 16$ unit directions u_j, the null removes at every token t the amount $|z_i(t)|\,\|w_i\|\,u_j$ (the legacy null removed one batch-averaged amount $\bar{a}\,W_d u_j$); u_j is $W_d^\top r_j$ normalized, with r_j a random unit code. The covariance-matched null instead uses $u_j \propto \sum_t g_{jt}(x_t - \bar{x})$ with $g_{jt} \sim \mathcal{N}(0,1)$ over the chunk's clean token activations x_t, a draw from a Gaussian with their sample covariance. Channel c clears if $\frac{1}{|R_i|}\sum_{r \in R_i} |\Delta_{i,r}^{(c)}| > q_{0.95}\{|\Delta_{u_j,r}^{(c)}| : j \le J, r \in R_i\}$. A null with no spread is reported as degenerate and not scored.

Transfer. The statistic is $\max_f \text{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k-sets, so the feature search is inside the null. The reverse leg scores the chosen feature's own top-k by the source activation. $p = (1 + \#\{\text{null} \ge \text{obs}\})/(1 + n)$; a registered claim uses $\max(p, p_{\text{rev}})$ (intersection–union), Holm across claims.

Shared-input agreement. On U, statistic (i) is the Spearman concordance of the two sides' signed level effects across series, and (ii) the cosine of their level-removed shape vectors, each channel in its own null units. Each side's floor ablates random feature sets matched to its activation on U and re-scores against the other side's measured effect; agreement exceeds both floors' $q_{0.95}$, "acts differently" falls below both $q_{0.05}$.

Minimum detectable effect. Before the private result is read, recentred per-series differences are shifted by a candidate effect and re-tested with the same bootstrap; bisection finds the smallest shift detected with power 0.8 (`analysis/power.py`).

Appendix D. Reproducibility

`examples/panel7_v2` holds the configuration, the report and every stage's JSON/parquet artifacts plus the confirmation (activations and SAE weights omitted for size); `paper/figures/make_figures.py --run DIR` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/panel7_v2.yaml` reproduces the run (add `--stages report` to re-render, and `--verify-provenance runs/panel7_v2_dev` to diff environments). The private split (v2 epoch 3, 971 series) was opened once (2026-10-02), so re-confirming requires a fresh epoch. Two events are disclosed. A first epoch minted for this look (v2 epoch 1) was rejected before any model saw it, because 716 of its 961 series equalled series of a split an earlier confirmation had consumed (epoch seeds depend only on the seed and epoch number). And the first confirm attempt crashed after every private battery finished, while assembling the claim ledger (a fixed family map lacked the failure-prediction family), before any statistic was written: `confirmation.json` did not exist, no private number was read, the registry hash and every frozen specification were unchanged, and the restart differed only by the crash fix. It is one test of the registry run twice for compute reasons; the artifact records `repeated_look: false`. The v3 run (`tsfm_lens/configs/panel7_v3.yaml`; dev 2,876 series, private epoch 1 of 1,024 series opened once on 2026-10-10, external 800 GIFT-Eval windows) records its confirmation in `runs/panel7_v3_dev/confirm/confirmation.json` (`repeated_look: false`); the step that re-ranks candidates by attribution is the separate, exploratory `configs/panel7_v3_sec42.yaml`, built on the same activations and dictionaries. In the v2 confirmation, 19 of the 21 non-confirmed single-feature claims carry a stored reading of "untestable" in `confirmation.json`; they were tested and not detected, the report recomputes the reading at render time, and no verdict is affected. Environment, as recorded in the run's `run_manifest.json`: Python 3.12.13, numpy 2.1.0, PyTorch 2.12.0 (CUDA 12.9 build), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7 (zarr 3 stores read back empty under the v2 API), pandas 2.3.3, scikit-learn 1.7.2; load-bearing pins and their reasons are in `DEPENDENCIES.md`. Tests: 2,647 (`tsfm_lens`) and 161 (`tsfm_benchmark`) on the release branch, run in continuous integration. Recorded stage wall-clock time on one RTX A5000: 75,900 s in total, of which `sae` 14,614 s and `concepts` 55,608 s (shared-input agreement 45,189 s); every other stage under 33 minutes.

Appendix E. Findings Ledger Index

Each identifier is an entry in `FINDINGS.md` giving exact numbers, models, evidence class, status, a reproduction pointer (configuration, command, artifact path), evidence class and status, and an interestingness score (1–5); negative results are included on equal terms. The table abridges each claim; evidence class and status are in the ledger. Prefixes: SH shared vs. unique; CA causal agreement; DE depth; PM per-model character; MP method positives; MN method negatives; BM benchmark trust.

Table 9: Ledger entries cited in this paper (score = interestingness, 1–5). *Section (§), table (T) or appendix (App.) of this paper where the result is reported.

ID	Claim (abridged)	In paper*	Score
SH-01	All 21 model pairs above shuffled-series nulls with concatenated windows on v2 and v3 dev and on a fresh sealed split (21/21)	§8, T8	5
SH-05	CKA and clustering AMI rank model pairs differently on three panels	T8	5
SH-06	Pair similarity splits into a geometric camp and a behavioral/causal camp; 7 metrics never pooled	T8	5
SH-10	Joint crosscoder loses to independent SAEs matched post hoc (0.2787 vs 0.3204, $p = 0.002$)	App. B	4
SH-16	No atlas concept spans the panel (at most 3 of 7 models; held out)	§8, T8	5
SH-18	Registered concept-transfer claims confirmed on private data: 19/20, 20/20 frozen, 20/20 on v2 epoch 0 (four models), 20/20 on v2 epoch 3 (seven models), 20/20 on v3 and on 800 raw real windows	§8, T8	5
SH-19	Unit-level correspondence fails rotation controls (0.148 → 0.331); participation ratio as low as 1.54	§8	5
SH-20	Per-target concepts mostly non-modular: 50 of 56 targets at 7 models	§4, §8	4
SH-23	Transfer and peak CKA barely change at windows 64 and 96; peak layer pair stable for 9 of 21 pairs	§8.1, App. B	3
SH-22	Atlas segregated by model; all 7 "same causal effect" verdicts are Chronos-2/Bolt; rung insensitive	§8, T8	5
CA-01	Perturbation-fingerprint agreement is pair-specific; TimesFM/Chronos-2 replicates on private data	T8	4
CA-03	Seasonality circuit: 1 head (TimesFM) vs. 5 non-additive heads (Chronos-T5-Base)	§8	4
CA-04	Attention mass at seasonal lags uncorrelated with causal importance ($\rho = 0.026$, -0.118)	§8, T8	4
CA-07	Median level share of a causal feature's effect 0.535; most features shape-causal beyond a level-removed null	§8	4
CA-10	Causal effect space is a continuum; families tile it but do not beat a shuffle null	§8, T8	4
CA-11	Held-out: 10/31 single-feature causal claims confirm; no cross-model one does (atlas 0/3, agreement 0/6); 43/82 in all	§8, T8	5
CA-12	v3 held-out: 41/63 claims; single-feature causal 11/24; transfer 20/20 private and 20/20 raw real	§8.1, T6	4
D-42C	Five attribution-ranked atoms carry a median 0.74 share of expressible forecast change; beat matched random sets on 30/31 targets	§8.5	3
DE-01	Whole-layer patching hides a depth-intensifying recency effect in the last pre-horizon window	§8	4
DE-03	Chronos-2 skip lens was a silent no-op (identical MASE at 12 layers); corrected reach 0.737 → 0.000	§4, §8	5
DE-04	Chronos-T5 encoder ends at depth 0.478 on a whole-stack axis; 50% comparable overlap with TimesFM	§4, §8	5
DE-05	Captured share of forecast FLOPs: 13.9% (Chronos-T5-Base) to 99.8% (Timer) at 7 models	§7, §8.6, T8	5
DE-08	Layer selection depends almost entirely on the selector (0/3 robust layers)	§8	4
PM-05	Chronos-2 most accurate and best observed; winners vary by family; held-out on v2 and on 7-model epoch 3	T8	4
PM-07	Models differ ∼5.4× in reversion to the mean; flat forecasts driven by weakly predictable data	§2	4
PM-09	Sampled models need identical seeds in compared calls; a mis-seeded null made Sundial look inert (1/71 → 12/71)	App. B	3
PM-11	Causally load-bearing TimesFM random-walk concept fails seed stability	§8	3
PM-13	Chronos-2 group attention unmeasured by design	§10	3
PM-17	Timer's SAE features do not clear the ablation null above chance (1.017×; 0 of 6 targets)	§8	3
MP-01	Token spans can be measured (IoU 1.0 on 6 checkpoints); zero-code adapter follows	§3	5
MP-02	Alignment gates must be normalized by their attainable ceiling	§3	3
MP-03	AuxK revival recovers dead SAE atoms 10–13× on real activations	§4	4
MP-04	Single-seed SAE gold ranking flips a bake-off verdict; 3 seeds fix it	§4	5
MP-05	SAE feature death set by layer geometry (exponent 1.02), not recipe	§10	5
MP-06	Ground-truth alignment of SAE features is real only over a permutation null	§8	3
MP-07	Ablation battery discriminates: 2.47× empirical chance at 7 models (Timer 1.02×, Bolt 5.70×)	§8	4
MP-09	Dev and private splits exchangeable (v2: energy $p = 0.951$, TOST 20/22; v3: $p = 0.995$, 22/22)	§7	3
MP-10	Effects survive mean, resample, clean ablation (69, 58, 49 of 91)	§8.2	3
MP-11	Token-level SAE training: Chronos-T5-Base token FVU 0.2835→0.0442, fewer dead features	§3.4, §10	4
MN-01	Similarity cannot detect lineage: untrained pair 0.878 > trained related pair 0.734	§8, T8	5
MN-07	Causal fingerprints are no cross-model matching key; effect vectors span ∼5.6 of 9 dimensions	§8	4
MN-14	0.835 of 927 prominent SAE features are causally null on private data (dev 0.810; 64.4% under the legacy null)	§8, T8	4
MN-15	SAE reconstruction moves the forecast 2.2× (v3 1.7×) more than removing a feature	§8, T8	4
MN-17	Generator-counterfactual concept mediation: 0/98 and 0/151	§8, T8	5
MN-18	Absence of agreement is not disagreement: 109/288 → 9	§4	4
MN-19	"Same inputs" must be top-k overlap, not whole-series correlation	§4	4
MN-20	Real Monash rows: lower effective dimension (1.76 vs 2.66) and more dead SAE atoms than the synthetic corpus	§10	3

ID	Claim (abridged)	In paper*	Score
MN-23	MASE denominator floors out on intermittent series	§4	3
MN-25	Zero-code adapter resolves 2 of 8 probed TSFM checkpoints	App. A	3
MN-28	Internals beat a free failure-prediction baseline only for the two weakest models; routing does not help	§8.7	4
MN-29	Legacy ablation null fails a planted-concept gate (FPR 0.168–0.248); profile-matched null passes ($\leq$0.003)	§6, T3	5
MN-30	Agreement rung is specific (0/9 false "same") but insensitive (0.4 and 0.0 of seeds)	§6, §8	5
MN-31	A planted decoy correlates with forecast error in 5/5 seeds but has no causal effect	App. B	3
MN-32	Subspace redesign of the agreement test fails its pre-fixed gate; planted distributed subspace does not clear a size-matched null	§10	4
MN-34	SAE fidelity does not predict causal clarity ($\rho = -0.326$, v3); 3$\times$ rows raise fidelity 0.749 $\rightarrow$ 0.82	§8.5, App. B	3
MN-36	Powered per-feature test (32 series, 64 draws) fails the planted gate (0.6); 8-series p95 rule passes (0.911)	§8.5	4
MN-37	Gradient-attribution candidates: no higher causal rate than random under the covariance null (0.085 vs 0.110)	§8.5	4
MN-38	Window-mean pooling understates TimesFM-vs-fine-token CKA by 0.30–0.43 (held-out); concatenation is lossless	§3.4, §8.3	5
BM-05	Counterfactual identity check caught a corruption-replay bug (244/298 $\rightarrow$ 0/1113)	§3	2
BM-06	42% of the dev corpus is weakly predictable real-derived data, driving the flat-forecast share	§10	4

References

J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, 2018.

T. Aksu, G. Woo, J. Liu, X. Liu, C. Liu, S. Savarese, C. Xiong, and D. Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

G. Alain and Y. Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (Workshop Track)*, 2017.

A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

A. F. Ansari, O. Shchur, J. Küken, A. Auer, B. Han, P. Mercado, S. S. Rangapuram, H. Shen, L. Stella, X. Zhang, M. Goswami, S. Kapoor, D. C. Maddix, P. Guerron, T. Hu, J. Yin, N. Erickson, P. M. Desai, H. Wang, H. Rangwala, G. Karypis, Y. Wang, and M. Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.

Y. Bansal, P. Nakkiran, and B. Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems*, 2021.

Y. Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1): 207–219, 2022.

N. Belrose, I. Ostrovsky, L. McKinney, Z. Furman, L. Smith, D. Halawi, S. Biderman, and J. Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023.

Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

J. Bloom, C. Tigges, A. Duong, and D. Chanin. SAELens. `https://github.com/jbloomAus/SAELens`, 2024.

T. Bolukbasi, A. Pearce, A. Yuan, A. Coenen, E. Reif, F. Viégas, and M. Wattenberg. An interpretability illusion for BERT, 2021.

T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. L. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.

A. C. Cameron, J. B. Gelbach, and D. L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008.

A. Das, W. Kong, R. Sen, and Y. Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 10148–10167, 2024.

F. Divo, M. Kraus, R. Härle, P. Ahrend, and K. Kersting. Finding the Zeitgeist in time series foundation models. In *ICLR 2026 Workshop on Time Series in the Age of Large Models*, 2026.

S. Dooley, G. S. Khurana, C. Mohapatra, S. Naidu, and C. White. ForecastPFN: Synthetically-trained zero-shot forecasting. In *Advances in Neural Information Processing Systems*, 2023.

C. Dwork, V. Feldman, M. Hardt, T. Pitassi, O. Reingold, and A. Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248):636–638, 2015.

B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1993.

N. Elhage, T. Hume, C. Olsson, N. Schiefer, T. Henighan, S. Kravec, Z. Hatfield-Dodds, R. Lasenby, D. Drain, C. Chen, R. Grosse, S. McCandlish, J. Kaplan, D. Amodei, M. Wattenberg, and C. Olah. Toy models of superposition. *Transformer Circuits Thread*, 2022.

J. Fiotto-Kaufman, A. R. Loftus, E. Todd, J. Brinkmann, K. Pal, D. Troitskii, M. Ripa, A. Belfki, C. Rager, C. Juang, A. Mueller, S. Marks, A. S. Sharma, F. Lucchetti, N. Prakash, C. Brodley, A. Guha, J. Bell, B. C. Wallace, and D. Bau. NNsight and NDIF: Democratizing access to open-weight foundation model internals. In *International Conference on Learning Representations*, 2025.

L. Gao, T. Dupré la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, and J. Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025.

P. Gao, E. Trautmann, B. Yu, G. Santhanam, S. Ryu, K. Shenoy, and S. Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*, page 214262, 2017.

A. Geiger, H. Lu, T. Icard, and C. Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, 2021.

A. Gelman and E. Loken. The statistical crisis in science. *American Scientist*, 102(6):460–465, 2014.

T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman, and P. Montero-Manso. Monash time series forecasting archive. In *Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 16115–16152, 2024.

S. Heimersheim and N. Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2733–2743, 2019.

S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2): 65–70, 1979.

R. Huben, H. Cunningham, L. R. Smith, A. Ewart, and L. Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations*, 2024.

M. Huh, B. Cheung, T. Wang, and P. Isola. Position: The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

R. J. Hyndman and A. B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006.

A. A. Ismail, M. Gunady, H. Corrada Bravo, and S. Feizi. Benchmarking deep learning interpretability in time series predictions. In *Advances in Neural Information Processing Systems*, 2020.

S. Jain and B. C. Wallace. Attention is not explanation. In *Proceedings of NAACL-HLT*, pages 3543–3556, 2019.

M. Kalnāre, S. Kitharidis, T. Bäck, and N. van Stein. Mechanistic interpretability for transformer-based time series classification. *arXiv preprint arXiv:2511.21514*, 2025.

Y. Kang, R. J. Hyndman, and F. Li. GRATIS: Generating time series with diverse and controllable characteristics. *Statistical Analysis and Data Mining*, 13(4):354–376, 2020.

S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 2023.

T. Kim, J. Kim, Y. Tae, C. Park, J.-H. Choi, and J. Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations*, 2022.

N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson. Captum: A unified and generic model interpretability library for PyTorch. *arXiv preprint arXiv:2009.07896*, 2020.

S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529, 2019.

N. Kriegeskorte, M. Mur, and P. Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008.

H. R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.

K. Lenc and A. Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 991–999, 2015.

Y. Li, J. Yosinski, J. Clune, H. Lipson, and J. Hopcroft. Convergent learning: Do different neural networks learn the same representations? In *International Conference on Learning Representations*, 2016.

J. Lindsey, A. Templeton, J. Marcus, T. Conerly, J. Batson, and C. Olah. Sparse crosscoders for cross-layer features and model diffing. *Transformer Circuits Thread*, Oct. 2024. URL `https://transformer-circuits.pub/2024/crosscoders/index.html`.

Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.

Y. Liu, H. Zhang, C. Li, X. Huang, J. Wang, and M. Long. Timer: Generative pre-trained transformers are large time series models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 32369–32399, 2024.

Y. Liu, G. Qin, Z. Shi, Z. Chen, C. Yang, X. Huang, J. Wang, and M. Long. Sundial: A family of highly capable time series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. arXiv:2502.00816.

C. H. Lubba, S. S. Sethi, P. Knaute, S. R. Schultz, B. D. Fulcher, and N. S. Jones. catch22: CAnonical time-series CHaracteristics: Selected through highly comparative time-series analysis. *Data Mining and Knowledge Discovery*, 33(6):1821–1852, 2019.

A. Makelov, G. Lange, A. Geiger, and N. Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024.

A. Makhzani and B. Frey. k-sparse autoencoders. *arXiv preprint arXiv:1312.5663*, 2013.

K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

A. Mishra. Dissecting Chronos: Sparse autoencoders reveal causal feature hierarchies in time series foundation models. In *ICLR 2026 Workshop on Time Series in the Age of Large Models*, 2026. arXiv:2603.10071.

A. S. Morcos, M. Raghu, and S. Bengio. Insights on representational similarity in neural networks with canonical correlation. In *Advances in Neural Information Processing Systems*, 2018.

N. Nanda and J. Bloom. TransformerLens. `https://github.com/TransformerLensOrg/TransformerLens`, 2022.

B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.

K. Oublal, Q. Bouniot, Q. Gan, S. Clémençon, and Z. Akata. TimeSAE: Causal sparse decoding for faithful explanations of black-box time series models. *arXiv preprint arXiv:2601.09776*, 2026.

G. Paulo and N. Belrose. Sparse autoencoders trained on the same data learn different features. *arXiv preprint arXiv:2501.16615*, 2025.

B. Phipson and G. K. Smyth. Permutation p-values should never be zero: Calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), 2010.

C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

M. Raghu, J. Gilmer, J. Yosinski, and J. Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, 2017.

K. Rasul, A. Ashok, A. R. Williams, H. Ghonia, R. Bhagwatkar, A. Khorasani, M. J. Darvishi Bayazi, G. Adamopoulos, R. Riachi, N. Hassen, M. Biloš, S. Garg, A. Schneider, N. Chapados, A. Drouin,

V. Zantedeschi, Y. Nevmyvaka, and I. Rish. Lag-Llama: Towards foundation models for probabilistic time series forecasting. *arXiv preprint arXiv:2310.08278*, 2023.

P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

H. Sakoe and S. Chiba. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1):43–49, 1978.

D. J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6): 657–680, 1987.

X. Shi, S. Wang, Y. Nie, D. Li, Z. Ye, Q. Wen, and M. Jin. Time-MoE: Billion-scale time series foundation models with mixture of experts. In *International Conference on Learning Representations*, 2025.

U. Simonsohn, J. P. Simmons, and L. D. Nelson. Specification curve analysis. *Nature Human Behaviour*, 4: 1208–1214, 2020.

G. J. Székely and M. L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.

A. Templeton, T. Conerly, J. Marcus, J. Lindsey, T. Bricken, B. Chen, A. Pearce, C. Citro, E. Ameisen, A. Jones, H. Cunningham, N. L. Turner, C. McDougall, M. MacDiarmid, A. Tamkin, E. Durmus, T. Hume, F. Mosconi, C. D. Freeman, T. R. Sumers, E. Rees, J. Batson, A. Jermyn, S. Carter, C. Olah, and T. Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL `https://transformer-circuits.pub/2024/scaling-monosemanticity/`.

J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, Y. Singer, and S. Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, 2020.

N. X. Vinh, J. Epps, and J. Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854, 2010.

S. Wan, M. Ma, H. Zhu, G. Liu, S. Wang, Q. Wen, and H. Liu. Forecast collapse in time-series foundation models. *arXiv preprint arXiv:2608.14106*, 2026.

M. Wiliński, M. Goswami, W. Potosnak, N. Żukowska, and A. Dubrawski. Exploring representations and interventions in time series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025.

G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 53140–53164, 2024.

Z. Wu, A. Geiger, A. Arora, J. Huang, Z. Wang, N. D. Goodman, C. D. Manning, and C. Potts. pyvene: A library for understanding and improving PyTorch models via interventions. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: System Demonstrations)*, pages 158–165, Mexico City, Mexico, 2024.

A. Yıldırım. Superposition is not necessary: A mechanistic interpretability analysis of transformer representations for time series forecasting. *arXiv preprint arXiv:2605.05151*, 2026.

F. Zhang and N. Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, 2024.

H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 2021.